

CHAPTER 4

PTYKES

Sur les cr dences, au salon vide: nul ptyx,
Aboli bibelot d’inanit  sonore,
Car le Ma tre est all  puiser des pleurs au Styx
Avec ce seul objet dont le N ant s’honore.

St phane Mallarm , “Sonnet en X”

Ptykes are algebraic objects defined by Girard to represent Π^1_{n+1} -formulas. Its name originated from the puzzling word ‘ptyx’ that appeared in St phane Mallarm ’s poem *Sonnet en X*. Girard defined n -ptykes as a functor from the category of $(n - 1)$ -ptykes to the category of well-orders preserving direct limits and pullbacks. 0-ptykes are well-orders, and 1-ptykes are dilators.¹ Girard and Ressayre analyzed basic properties of n -ptykes in [GirardRessayre1985], which is written in French, which is unfortunate for non-French speakers.

In this chapter, we develop the theory of 2-ptykes. Most literature calls 2-ptykes just ptykes, but we reserve the word ptyx for general objects. The reader could find that most arguments in ?? and ?? carry over in a more complex form. One reason for the complexity is that we simultaneously use semidilators and semidendrograms, so we need to convert one into the other. Another reason is that semidilators (or better to say, semidendrograms) have a tree structure, and we often need to consider ‘hidden’ terms to convert proofs for dilators to proofs for ptykes. I could not complete every detail on general ptykes in this dissertation, but at least for n -ptykes, most basic facts about 2-ptykes also hold for n -ptykes *mutatis mutandis*. Many ideas of the first three sections are from

¹I was asked why n -ptykes are not named as n -dilators. I would agree with this suggestion, but it is too late to change the terminology. For future terminologies, we should avoid making new words for the ptyx-version of dilator notions. For example, a 2-ptyx version of flower should be 2-flower, and should not use a new word.

[GirardRessayre1985], and [Section 4.4](#) is influenced by Freund [Freund2021Pi14].

4.1 Denotation system of 2-semiptykes

Definition 4.1.1. A 1-arity diagram $\mathbb{D}$ is a commutative diagram over the category of natural semidendrogramoids with *semidendrogram quasimorphisms* of the form

$$\mathbb{D} = \begin{array}{ccc} a_{\cap} & \xrightarrow{f_{\cap,1}} & a_1 \\ f_{\cap,0} \downarrow & & \downarrow f_{1,\cup} \\ a_0 & \xrightarrow{f_{0,\cup}} & a_{\cup} \end{array} \quad (4.1)$$

such that both $f_{0,\cup}$ and $f_{1,\cup}$ are morphisms (so they preserve terminal points), the above diagram is a pullback, and $\text{ran } f \cup \text{ran } g = a_{\cup}$. We say a 1-arity diagram is *trivial* if $a_{\cap} = a_0 = a_1 = a_{\cup}$. For a 1-arity diagram of the form (4.1), the diagram $-\mathbb{D}$ is

$$-\mathbb{D} = \begin{array}{ccc} a_{\cap} & \xrightarrow{f_{\cap,0}} & a_0 \\ f_{\cap,1} \downarrow & & \downarrow f_{0,\cup} \\ a_1 & \xrightarrow{f_{1,\cup}} & a_{\cup} \end{array}$$

As we did in ??, we understand a 1-arity diagram as a functor from the category $\mathbb{L}_2$ to the category of natural semidendrogramoids with semidendrogram quasimorphisms.

Remark 4.1.2. The condition ‘ $f_{\cap,0}$ and $f_{\cap,1}$ are semidendrogram quasimorphisms’ in [Definition 4.1.1](#) cannot be fortified by ‘ $f_{\cap,0}$ and $f_{\cap,1}$ are semidendrogram morphisms,’ otherwise we cannot consider the term configurations of two finite semidilators with their overlapping parts hidden, like illustrated in [Example 3.3.9](#).

We also use the following notation as we did in ??:

$$\mathbb{D} = \begin{array}{ccc} \mathbb{D}(\perp) & \xrightarrow{\quad} & \mathbb{D}(1) \\ \downarrow & & \downarrow \mathbb{D}(1,\top) \\ \mathbb{D}(0) & \xrightarrow{\mathbb{D}(0,\top)} & \mathbb{D}(\top) \end{array}$$

Definition 4.1.3. Let $a_0, \dots, a_{m-1}$ be natural semidendrograms and let $\mathbb{L}_m$ be the free distributive lattice generated by $\{0, 1, \dots, m-1\}$. A *1-IU diagram* $\mathbb{A}$ for $a_0, \dots, a_{m-1}$ is a collection of objects $\{a_i \mid i \in \mathbb{L}_m\}$ and semidendrogramoid quasimorphisms $f_{ij}: a_i \rightarrow a_j$ for $i \leq j$ such that the following holds:

1. f_{ii} is the identity map.
2. For $i \leq j \leq k$, $f_{ik} = f_{jk} \circ f_{ij}$.
3. For each $i, j \in \mathbb{L}_m$, the following diagram is a pullback and $\text{ran } f_{i, i \vee j} \cup \text{ran } f_{j, i \vee j} = a_{i \vee j}$:

$$\mathbb{A}[i, j] = \begin{array}{ccc} a_{i \wedge j} & \xrightarrow{f_{i \wedge j, j}} & a_j \\ \downarrow f_{i \wedge j, i} & & \downarrow f_{j, i \vee j} \\ a_i & \xrightarrow{f_{i, i \vee j}} & a_{i \vee j} \end{array}$$

Moreover, $f_{i, i \vee j}$ and $f_{j, i \vee j}$ are semidendrogram morphisms.

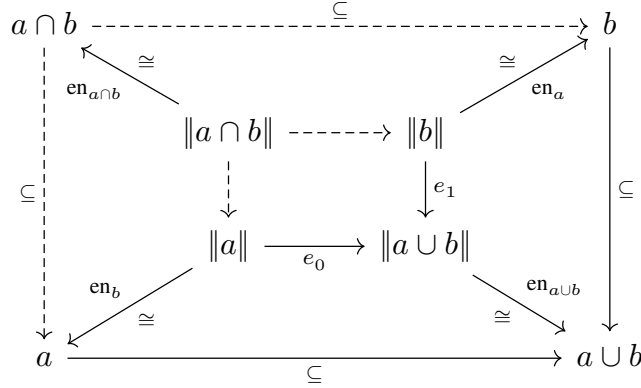
Alternatively, we can think of an IU diagram as a functor from $\mathbb{L}_n$ as a category to the category of natural semidendrogramoids with quasimorphisms.

Definition 4.1.4. Let $a, b \subseteq C$ be two finite sub-semidendrograms of C . The *1-diagram* $\text{Diag}_C(a, b)$ of a, b over C is the unique 1-arity diagram isomorphic to the inclusion diagram

$$\begin{array}{ccc} a \cap b & \xrightarrow{\subseteq} & b \\ \downarrow \subseteq & & \downarrow \subseteq \\ a & \xrightarrow{\subseteq} & a \cup b \end{array}$$

Here note that $a \cap b$ may *not* be a sub-semidendrogram of C since a terminal point of $a \cap b$ may not be a terminal of C . More precisely, $\text{Diag}_C(a, b)$ is the innermost diagram

in the below commutative diagram:



In general, for finite semidendrograms $a_0, \dots, a_{n-1} \subseteq C$ of a semidendrogram C , $\text{Diag}_C(a_0, \dots, a_{n-1})$ is the unique IU diagram isomorphic to the inclusion diagram $\mathcal{J}: \mathbb{L}_n \rightarrow \mathcal{P}(a_0^* \cup \dots \cup a_{n-1}^*)$, where $\mathcal{J}$ can be thought of a lattice homomorphism from $(\mathbb{L}_n, \wedge, \vee)$ to $(\mathcal{P}(a_0^* \cup \dots \cup a_{n-1}^*), \cap, \cup)$ satisfying $\mathcal{J}(i) = a_i$ for every $i < n$.

Definition 4.1.5. A 2-semiptyx is a set P of P -terms t with a natural semidendrogram $\text{arity}(t)$. For two P -terms t_0, t_1 and a 1-arity diagram

$$\mathbb{A} = \begin{array}{ccc} a_{\cap} & \dashrightarrow & a_1 \\ \downarrow & & \downarrow \\ a_0 & \longrightarrow & a_{\cup} \end{array}$$

such that $a_i = \text{arity } t_i$, we have a relation $t_0 <_{\mathbb{A}} t_1$. We call $\mathbb{A}$ a 1-arity diagram for t_0 and t_1 . P also satisfies the following:

1. (Irreflexivity) If $t_0 = t_1$ and $\mathbb{A}$ is trivial, then $t_0 <_{\mathbb{A}} t_0$ does not hold.
2. (Linearity) If $t_0 \neq t_1$ or $\mathbb{A}$ is not trivial, then one of $t_0 <_{\mathbb{A}} t_1$ or $t_1 <_{-\mathbb{A}} t_0$ must hold.
3. (Transitivity) For three P -terms t_0, t_1, t_2 such that $\text{arity}(t_i) = a_i$, and an IU diagram $\mathbb{A}$ for t_0, t_1, t_2 , if $t_0 <_{\mathbb{A}[0,1]} t_1$ and $t_1 <_{\mathbb{A}[1,2]} t_2$, then $t_0 <_{\mathbb{A}[0,2]} t_2$.

We say P is *positive* if $\text{arity}(t)$ is a natural predendrogram for every P -term t .

Like semidilators, we can first-order axiomatize the definition of 2-semiptykes over the multi-sorted language. Also, we have the following:

Proposition 4.1.6. *Let P be a 2-semiptyx, $t_0, t_1 \in \text{field}(P)$, and $\mathbb{D}$ a 1-arity diagram for t_0, t_1 . Then neither $t_0 <_{\mathbb{D}} t_1$ nor $t_1 <_{-\mathbb{D}} t_0$ both hold.*

Proof. Suppose not, assume that $t_0 <_{\mathbb{D}} t_1 <_{-\mathbb{D}} t_0$. Assume that $\mathbb{D}$ has of the form (4.1), and consider the 1-IU diagram $\mathbb{D} = \text{Diag}_{a_0}(\text{ran } f, \text{ran } g, \text{ran } f)$. Then we have $\mathbb{D}[0, 1] = \mathbb{D}$, $\mathbb{D}[1, 2] = -\mathbb{D}$, and $\mathbb{D}[0, 2]$ is the trivial diagram for a_0 . By transitivity of a 2-semiptyx, we have $t_0 <_{\mathbb{D}[0, 2]} t_0$, so $t_0 <_{\mathbb{K}} t_0$ for a trivial diagram $\mathbb{K}$, contradicting with the irreflexivity. $\square$

We can also talk about the subobject of a 2-semiptyx:

Definition 4.1.7. Let P be a 2-semiptyx and $X \subseteq \text{field}(P)$. A *sub-2-semiptyx* $P \upharpoonright X$ induced by X is a 2-semiptyx whose field is X with the same arity function and comparison rules for P .

4.2 Category-theoretic aspects of 2-semiptykes

Semidilators can be thought of as functors from LO to LO. Likewise, 2-semiptykes should be functors from the category SDil of semidilators to LO. However, unlike semidilators in which arities and arity diagrams are defined in terms of finite linear orders, we defined 2-semiptykes with finite semidendrograms and not finite semidilators. It makes the technical definition of 2-semiptykes slightly different from our expectation:

Definition 4.2.1. From a given 2-semiptyx P and a semidendrogram C , let us define $P(C)$, the *application of P to C* by

$$P(C) = \{(t, a) \mid t \text{ is a } P\text{-term, } a \subseteq C, \text{ and } \|a\| = \text{arity } t\}. \quad (4.2)$$

We write $t(a)$ instead of (t, a) . The order of $P(C)$ is given by

$$P(C) \models s(a) < t(b) \iff P \models s <_{\text{Diag}_C(a,b)} t$$

For a semidendrogram morphism $f: C \rightarrow C'$, consider the map $P(f): P(C) \rightarrow P(C')$ given by $P(f)(t, a) = (t, f[a])$.

Proposition 4.2.2. *Let P be a 2-semiptyx and C be a semidendrogram. Then $P(C)$ is a linear order. Also, P induces a functor $\text{SDen} \rightarrow \text{LO}$.*

Proof. Let us prove that $P(C)$ is a linear order. The irreflexivity and the linearity of a semidilator imply that the comparison relation over $P(C)$ satisfies the irreflexivity and the linearity, respectively. To see the transitivity of $<_{P(C)}$, suppose that we have $P(C) \models s_0(a_0) < s_1(a_1) < s_2(a_2)$. Now let $\mathbb{A}$ be the 1-IU diagram isomorphic to a diagram induced by intersections and unions of a_0, a_1, a_2 , and the insertion maps. In particular, $\mathbb{A}(i, j)$ is the induced diagram $\text{Diag}_C(a_i, a_j)$ from a_i and a_j for $i \neq j < 2$. Then we have

$$s_0 <_{\mathbb{A}(0,1)} s_1 \wedge s_1 <_{\mathbb{A}(0,2)} s_2.$$

Thus we have $s_0 <_{\mathbb{A}(0,2)} s_2$ by the transitivity of a semidilator, which implies $P(C) \models s_0(a_0) < s_2(a_2)$. To see that P induces a functor, it suffices to show that the following holds:

1. $P(\text{Id}_C) = \text{Id}_{P(C)}$,
2. $P(f \circ g) = P(f) \circ P(g)$.

Both of them are easy to verify, so we leave their proof to the reader. $\square$

Definition 4.2.3. Let P and Q be two 2-semiptyxes. A map $\iota: P \rightarrow Q$ is an *embedding* or a *2-semiptyx morphism* if it satisfies the following conditions:

1. ι is a function field P to field Q .

2. ι preserves the arity: i.e., $\text{arity}(\iota(t)) = \text{arity}(t)$ for every $t \in \text{field } P$,
3. For each two terms $t_0, t_1 \in \text{field } P$ and a 1-arity diagram $\mathbb{D}$ between them, we have $t_0 <_{\mathbb{D}} t_1$ iff $\iota(t_0) <_{\mathbb{D}} \iota(t_1)$.

An embedding ι is an *isomorphism* if **whenever** $\iota: \text{field } P \rightarrow \text{field } Q$ is a bijection and ι^{-1} is also an embedding. We denote $P \cong Q$ if there is an isomorphism between P and Q . We also write $P \leq Q$ if there is an embedding from P to Q .

Like the semidilator case, a 2-semiptyx morphism is also one-to-one, and a bijective embedding is an isomorphism:

Lemma 4.2.4. *If $\iota: P \rightarrow Q$ is an embedding, then $\iota: \text{field } P \rightarrow \text{field } Q$ is injective.*

Proof. Suppose the contrary that we have two different P -terms t_0 and t_1 such that $\iota(t_0) = \iota(t_1)$. Then $\text{arity } t_0 = \text{arity } \iota(t_0) = \text{arity } t_1$, and let $\mathbb{D}$ be the trivial diagram for t_0 . Then either $t_0 <_{\mathbb{D}} t_1$ or $t_1 <_{\mathbb{D}} t_0$ holds, so we get $\iota(t_0) <_{\mathbb{D}} \iota(t_1)$ or $\iota(t_1) <_{\mathbb{D}} \iota(t_0)$. It contradicts the irreflexivity of Q . $\square$

Lemma 4.2.5. *If $\iota: P \rightarrow Q$ is an embedding and a bijection, then $\iota^{-1}: Q \rightarrow P$ is an embedding.*

Proof. Suppose that $Q \models t_0 <_{\mathbb{D}} t_1$. Then we can find $s_0, s_1 \in \text{field}(P)$ such that $\iota(s_0) = t_0$ and $\iota(s_1) = t_1$. If $P \models s_1 <_{-\mathbb{D}} s_0$, then $E \models t_1 <_{-\mathbb{D}} t_0$, so $Q \models t_1 <_{-\mathbb{D}} t_0 <_{\mathbb{D}} t_1$, which is impossible. Hence $P \models s_0 <_{\mathbb{D}} s_1$. $\square$

We can view an embedding between two semidilators as a natural transformation in the following manner:

Definition 4.2.6. For a 2-semiptyx morphism $\iota: P \rightarrow Q$ and a semidendrogram C , define $\iota_D: P(C) \rightarrow Q(C)$ by

$$\iota_D(t, a) = (\iota(t), a).$$

Proposition 4.2.7. *Let $\iota: P \rightarrow Q$ be 2-semiptyx morphism between two 2-semiptykes P and Q . Then $\iota_C: P(C) \rightarrow Q(C)$ is a strictly increasing map. Furthermore, if $f: C \rightarrow C'$ is a semidendrogram morphism from C to C' , then the following diagram commutes:*

$$\begin{array}{ccc} P(C) & \xrightarrow{P(f)} & P(C') \\ \iota_C \downarrow & & \downarrow \iota_{C'} \\ Q(C) & \xrightarrow{Q(f)} & Q(C') \end{array}$$

Proof. For ι_C being strictly increasing, suppose that we have $P(C) \models s(a) < t(b)$, and let $\mathbb{D}$ be the induced diagram from a and b . Then we have $s <_{\mathbb{D}} t$, so $\iota(s) <_{\mathbb{D}} \iota(t)$. Since $\text{arity } s = \text{arity } \iota(s)$ and $\text{arity } t = \text{arity } \iota(t)$, $(\iota(s), a), (\iota(t), b)$ are in $Q(C)$ and moreover we have $Q(C) \models \iota(s)(a) < \iota(t)(b)$, which is $Q(C) \models \iota_C(s(a)) < \iota_C(t(b))$. The commutativity is easy to check. $\square$

4.3 2-Preptykes and 2-ptykes

In this section, we define 2-preptykes and 2-ptykes. Girard's original definition of 2-ptykes is a functor from the category Dil of dilators to the category of well-orders preserving direct limits and pullbacks, and 2-preptykes are a direct limit of 2-ptykes. Every Girard-styled 2-ptyx extends to a functor from the category PreDil of predilators to LO , and we can further extend it by 'ignoring' non-predilator parts. Although we will not follow Girard's original definition, we follow some spirit of his definition; i.e., we will always require 2-preptykes and 2-ptykes to be *positive*, so they ignore non-predilator parts.

Definition 4.3.1. A 2-semiptyx P is *positive* if for every $t \in \text{field}(P)$, $\text{arity}(t)$ is a predendrogram.

Definition 4.3.2. Let us say a 1-arity diagram

$$\mathbb{D} = \begin{array}{ccc} a_{\cap} & \dashrightarrow & a \\ \downarrow & & \downarrow e_1 \\ a & \xrightarrow{e_0} & a_{\cup} \end{array} \quad (4.3)$$

is *monotone* if a is a natural predendrogram and $e_0(x) \leq_{\text{KB}} e_1(x)$ for every $x \in a$, where $x \leq_{\text{KB}} y$ if and only

1. Either $x = y$, or
2. If m is the least number such that $x \upharpoonright m \neq y \upharpoonright m$, we have $a \models x \upharpoonright m < y \upharpoonright m$.

A 2-semiptyx P is *monotone* if it satisfies the *monotonicity condition*: For every $t \in \text{field}(P)$ and a non-trivial monotone 1-arity diagram $\mathbb{D}$ of the form (4.3), $t <_{\mathbb{D}} t$ holds. A positive monotone 2-semiptyx P is called a *2-preptyx*.

Like predilators, we can formulate the definition of 2-preptyx in a multi-sorted first-order language of 2-semiptykes. Also, we can formulate a 2-preptyx in a structural manner: In the following proposition, $f \leq g$ for two 2-preptyx morphisms $f, g: C \rightarrow C'$ means for every $x \in C$, $f(x) \leq_{\text{KB}} g(x)$.

Proposition 4.3.3. *A positive 2-semiptyx P is a 2-preptyx iff for every predendrogram morphisms $f, g: C \rightarrow C'$, if $f \leq g$ then $P(f) \leq P(g)$.*

Proof. For the left-to-right direction, suppose that $f, g: C \rightarrow C'$ are two predendrogram morphisms such that $f \leq g$. We claim that for each $x \in P(C)$, $P(f)(x) \leq P(g)(x)$. Every element of $P(C)$ has the form $t(a)$ for some $t \in \text{field}(P)$ and $a \subseteq C$ such that $\|a\| = \text{arity } t$. Hence

$$P(C') \models P(f)(t(a)) \leq P(g)(t(a)) \iff P(C') \models t(f[a]) \leq t(g[a]).$$

If $f[a] = g[a]$, then $P(C') \models t(f[a]) \leq t(g[a])$ clearly holds. Now consider the case $f[a] \neq g[a]$: Let $c = \text{arity } t$ and $\mathbb{D} = \text{Diag}_C(f[a], g[a])$. Then $\mathbb{D}$ is not trivial. Moreover,

if $\mathbb{D}$ has the form (4.3), then for $x \in a$,

$$e_0(x) \leq_{\text{KB}} e_1(x) \iff \text{en}_{f[a]}(x) \leq_{\text{KB}} \text{en}_{g[a]}(x).$$

The latter inequality holds by the assumption $f \leq g$ and the fact that $\text{en}_{f[a]} = f \circ \text{en}_a$. This shows $e_0 \leq e_1$, so $\mathbb{D}$ is monotone. Since P is monotone, $P \models t <_{\mathbb{D}} t$. Therefore $P(C) \models t(f[a]) < t(g[a])$.

For the right-to-left direction, let $\mathbb{D}$ be a non-trivial monotone 1-arity diagram of the form (4.3), and t be a P -term of arity a . Then $e_0 \leq e_1$ implies $P(e_0) \leq P(e_1)$. In particular,

$$P(a_{\cup}) \models P(\text{Dec}(e_0))(t(a)) \leq P(\text{Dec}(e_1))(t(a)).$$

Observe that $P(e_i)(t(a)) = t(\text{ran } e_i)$, so we get

$$P(a_{\cup}) \models t(\text{ran } e_0) \leq t(\text{ran } e_1).$$

By the non-triviality of $\mathbb{D}$, we have $\text{ran } e_0 \neq \text{ran } e_1$, so the inequality is strict. Since the induced diagram from $\text{ran } e_0$ and $\text{ran } e_1$ is again $\mathbb{D}$, we get $P \models t <_{\mathbb{D}} t$. $\square$

Definition 4.3.4. A positive 2-semiptyx P is a 2-ptyx if for every dendrogram C , $P(C)$ is a well-order.

The following proposition follows the proof of [Proposition 1.3.5](#). However, we work with predendrograms instead of linear orders, so the proof is more convoluted:

Proposition 4.3.5. *If P is a 2-ptyx, then P is a 2-preptyx.*

More precisely, a positive 2-semiptyx P is a 2-preptyx if $P(C)$ is well-founded for every locally well-founded predendrogram C of finite height such that for every $u \in C$, the ordertype of the set of immediate successors of u is less than $\omega \cdot 2$.

Proof. Suppose that P is a positive 2-semiptyx that is not a 2-preptyx. Then we have a non-trivial monotone 1-arity diagram $\mathbb{D}$ of the form (4.3) and a P -term t such that

arity $t = a$. Similar to the proof of [Proposition 1.3.5](#), we decompose $\mathbb{D}$ to simpler 1-arity diagrams.

Let $\{u_i \mid i < N\} \subseteq a$ be the set of $u \in a$ such that $e_0(u) < e_1(u)$. We enumerate $\{u_i \mid i < N\}$ in a way that if $i < j < N$, then

1. $\text{lh } u_i \leq \text{lh } u_j$.
2. If u_i and u_j have the same immediate predecessor, then $u_i < u_j$.

If $e_0(x) <_{\text{KB}} e_1(x)$, then x is an extension of some u_i . In fact, if m is the least number satisfying $e_0(x) \upharpoonright m < e_1(x) \upharpoonright m$, then we can take $u_i = x \upharpoonright m$. (Note that $e_0(x) \upharpoonright m = e_0(x \upharpoonright m)$ and the same for e_1 .) For each $i \leq N$, define $f_i: a \rightarrow a_{\cup}$ by

$$f_i(x) = \begin{cases} e_1(x) & x \text{ extends } u_j \text{ for some } j \geq N - 1 - i, \\ e_0(x) & \text{Otherwise.} \end{cases}$$

For a technical convenience, we assume u_{-1} is extended to every node. Then $f_0 = e_0$ and $f_N = e_1$.

Claim 4.3.6. Each f_i is a predendrogram morphism.

Proof. It suffices to show that if y is extended to u_j for some $j \geq N - 1 - i$, then $e_0(y) = e_1(y)$. In fact, the definition of $\{u_j \mid j < N\}$ implies that $e_0(u_j)$ and $e_1(u_j)$ have the same immediate predecessor, so $e_0(u_j \upharpoonright m) = e_1(u_j \upharpoonright m)$ for every $m < \text{lh } u_j$. $\square$

Moreover, $f_i(x) \neq f_{i+1}(x)$ iff x extends u_{N-1-i} , and in this case, we have $f_i(x) <_{\text{KB}} f_{i+1}(x)$.

Now consider the IU diagram $\mathbb{D} = \text{Diag}_{a_{\cup}}(\text{ran } f_0, \text{ran } f_1, \dots, \text{ran } f_N)$. Then $\mathbb{D}[0, N] = \mathbb{D}$, so the transitivity of a 2-preptyx implies there is $i < N$ such that $P \models t \not\prec_{\mathbb{D}[i, i+1]} t$. Now let us write $\mathbb{K} = \mathbb{D}[i, i+1]$. If we assume $\mathbb{K}$ takes the

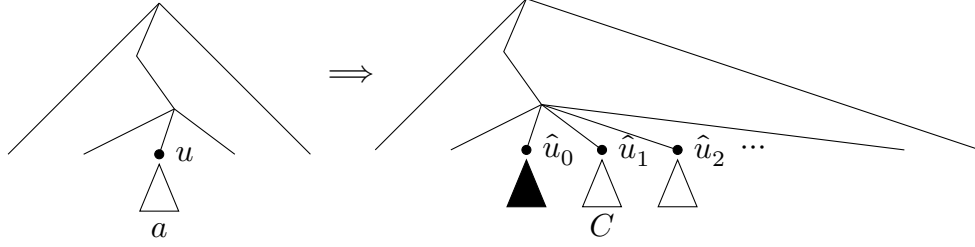


Figure 4.1: The construction of C from a . The filled triangle, $\hat{u}_0$, and the other uncopied part constitute a_0 . Here, $\hat{u}_i = \omega \cdot u +_u i$.

form

$$\mathfrak{K} = \begin{array}{ccc} a'_\cap & \dashrightarrow & a \\ \downarrow & & \downarrow g_1 \\ a & \xrightarrow{g_0} & a'_\cup \end{array}$$

then $g_0(x) \leq_{\text{KB}} g_1(x)$ for every $x \in a$, and they differ iff x extends $u := u_{N-1-i}$.

Now let us build a locally well-founded predendrogram C of finite height with a sequence of subpredendrograms $\{a_i \mid i \in \mathbb{N}\}$ such that $\text{Diag}_C(a_i, a_j) = \mathfrak{K}$ for $i < j$. If we have such C and a_i , then we have

$$P(C) \models t(a_0) > t(a_1) > t(a_2) > \dots,$$

so $P(C)$ is ill-founded. The basic idea of the construction of C is ‘copying’ parts below u from a ω many times; See [Figure 4.1](#) for the illustration of the construction.

Let us start from a natural predendrogram a . We first replace every component of a with a multiple of ω , that is, we consider a new predendrogram a' such that the map

$$\langle e_0, e_1, \dots, e_{l-1} \rangle \mapsto \langle \omega \cdot e_0, \omega \cdot e_1, \dots, \omega \cdot e_{l-1} \rangle$$

is an isomorphism $a \rightarrow a'$. We denote this map by $x \mapsto \omega \cdot x$. For $x \in a$ extending u , we define a new tuple

$$\omega \cdot x +_u i = \langle \omega \cdot x_0, \dots, \omega \cdot x_{l-2}, (\omega \cdot x_{l-1} + i), \omega \cdot x_l, \dots, \omega \cdot x_{m-1} \rangle, \quad (4.4)$$

where $x = \langle x_0, \dots, x_{m-1} \rangle$ and $u = \langle x_0, \dots, x_{l-1} \rangle$ for $l \leq m$. Then let us consider the

new predendrogram $a'' = a' \cup \{\omega \cdot x +_u i \mid i < \omega \wedge u \multimap^* x\}$ with the following additional information:

1. If $a \models v \multimap u$, then $a'' \models \omega \cdot v \multimap \omega \cdot u +_u i$. For $x, y \in a$ extending u such that $a \models x \multimap y$, $a'' \models \omega \cdot x +_u i \multimap \omega \cdot y +_u j$.
2. $a'' \models \omega \cdot u +_u i < \omega \cdot u +_u j$ iff $i < j$.
3. $\pi^{a''}(\omega \cdot x +_u i) = \pi^a(x)$.

Then consider the subpredendrogram $a_i \subseteq a''$ given by

$$a_i = \{\omega \cdot x \mid u \not\multimap^* x\} \cup \{\omega \cdot x + i \mid u \multimap^* x\}. \quad \square$$

Conversely, we can show that every *finite* 2-preptyx is a 2-ptyx. Let us start with counting the number of 1-arity diagrams for two copies of a predendrogram:

Lemma 4.3.7. *For a natural predendrogram a , there are finitely many 1-arity diagrams between a and a .*

Proof. We may calculate the precise number of the 1-arity diagrams between a and a , but we only provide a dull upper bound. We will first find a finite family of natural predendrograms embedding two copies of a , then count the number of such families. A natural dendrogram C is *h -short N -branching* if for $u \in C$

1. $\text{lh } u \leq h$, and
2. Either u is a terminal point or $u \frown \langle i \rangle \in C$ for every $i < N$.

We first set the upper bound of the number of h -short N -branching natural predendrograms. Let us imagine a forest of height h , with each node branching N times until it reaches level h . We color the edges of each level: In Level 0, we color 0 or ‘terminates,’ which means we will ignore successors following the edge. In Level 1, we color the

edge in either 01, 10, or ‘terminates,’ Similarly, in Level k , we color the edge in an appropriate permutation of $(k + 1)$ or ‘terminates.’ Some of the colorings give an h -short N -branching natural predendrograms, so we have at most

$$(1! + 1)^N + (2! + 1)^{2N} + \dots + (h! + 1)^{hN}$$

many h -short N -branching natural predendrograms. We can also see that for a 1-arity diagram $\mathbb{D}$ of the form (4.3), if we take

- N to be the largest number of immediate successors of a node of C , and
- h is the maximum length of a node of C ,

then $a_{\cup}$ embeds to a h -short N -branching natural dendrogram. Hence, the number of 1-arity diagrams of the form (4.3) is less than or equal to the number of pairs of subsets of an h -short N -branching natural dendrogram, which is at most

$$(2^{(1!+1)^N + (2!+1)^{2N} + \dots + (h!+1)^{hN}})^2.$$

□

The proof of the next proposition uses Ramsey’s theorem for pairs, so it is not provable over RCA_0 or ECA_0 :

Proposition 4.3.8. *Every finite 2-preptyx is a 2-ptyx. In fact, if P is a finite 2-preptyx, then $P(C)$ is a well-order for every locally well-founded predendrogram C .*

Proof. Suppose that P is a finite positive 2-semiptyx that is not a 2-ptyx. We claim that P is not a 2-preptyx. Fix a locally well-founded predendrogram C such that $P(C)$ is ill-founded, so we have a sequence $\langle t_i \mid i \in \mathbb{N} \rangle$ of P -terms and finite subpredendrograms a_i of C such that

$$P(C) \models t_0(a_0) > t_1(a_1) > t_2(a_2) > \dots.$$

By the pigeonhole principle, we may assume every t_i is equal to some t . Then consider the coloring F of domain $[\mathbb{N}]^2$ given by $F(i, j) = \text{Diag}_C(a_i, a_j)$. The range of F is finite

by [Lemma 4.3.7](#), so by Ramsey's theorem for pairs, we have an infinite homogeneous set $H \subseteq \mathbb{N}$ with value $\mathbb{D}$.

For $i, j \in H$, $P(C) \models t(a_i) > t(a_j)$ implies $P \models t \not\prec_{\mathbb{D}} t$. $\mathbb{D}$ is not trivial, otherwise $a_0 = a_1 = a_2 = \dots$, contradicting with $t(a_0) > t(a_1) > t(a_2) > \dots$. Now we claim that $\mathbb{D}$ witnesses P is not a 2-preptyx: Suppose that $\mathbb{D}$ has the form (4.3). We claim that $e_0 \leq e_1$ holds: Otherwise, we have $x \in a$ such that $e_0(x) \succ_{\text{KB}} e_1(x)$. Since $\text{Diag}_C(a_i, a_j) = \mathbb{D}$, we also have $\text{en}_{a_i}(x) \succ_{\text{KB}} \text{en}_{a_j}(x)$ for every $i < j$. In particular, we have $\prec_{\text{KB}}$ -descending sequence $\langle \text{en}_{a_i}(x) \mid i \in \mathbb{N} \rangle$ over C of fixed length, i.e., the length of $\text{en}_{a_i}(x)$ is constant. This is impossible since C is locally well-founded. $\square$

Remark 4.3.9. There is a non-trivial non-positive finite 2-semiptyx P such that $P(C) = 0$ for every predendrogram C . An easy example of P is when P has a single term of arity opp . Hence the positive condition of a 2-preptyx is essential to make the theory of 2-preptykes more similar to that of predilators.

4.4 2-Semiptykes as functors with support transformation

In this section, we provide an alternative formulation of 2-semiptykes close to the Freund-style definition of semidilators.

Definition 4.4.1. An *F-2-semiptyx* is a pair (P, supp) , where $P: \text{SDen} \rightarrow \text{LO}$ is a functor and $\text{supp}: P \rightarrow [\cdot]^{<\omega}$ is a natural transformation satisfying the *support condition*

$$\{\sigma \in P(C) \mid \text{supp}_C(\sigma) \subseteq \text{ran}(f)\} \subseteq \text{ran}(F(f)) \quad (4.5)$$

for every semidendrogram morphism $f: C \rightarrow C'$. We often omit supp and simply write P to mean (P, supp) .

To formulate the 2-preptyx à la Freund, we need a way to describe ‘ignoring’ the non-predendrogram part of a semidendrogram:

Definition 4.4.2. Let C be a semidendrogram and $x \in C$. We say x is *positive* if $\wp_t = \text{lh}(t)$. We define $\text{pos}(C)$ to be the set of $x \in C$ such that x extends to a positive terminal node of C .

We can easily see that $\text{pos}(C)$ forms a sub-predendrogram of C . Now, let us state the definition of a 2-preptyx with support transformation:

Definition 4.4.3. A F-2-semiptyx P is a *F-2-preptyx* if it satisfies

1. P is *positive*, i.e., For every semidendrogram C , $P(C) = P(\text{pos}(C))$.
2. P is *monotone*, i.e., $f \leq g$ implies $P(f) \leq P(g)$ for every predendrogram morphism $f, g: C \rightarrow C'$.

Definition 4.4.4. The *category of F-2-semiptyxes* $\text{SPtyx}_{\mathbb{F}}^2$ is the category whose objects are F-2-semiptyxes, and whose morphisms are natural transformations.

Like the semidilator case, every natural transformation between F-2-semiptyxes preserves support transformation:

Proposition 4.4.5. Suppose that P, Q are F-2-semiptyxes and $\epsilon: P \rightarrow Q$ is a natural transformation. Then $\text{supp}^Q = \iota \circ \text{supp}^P$.

Proof. Let C be a semidendrogram and $\sigma \in P(C)$. We also let $\epsilon_\sigma: \text{supp}_C^P(\sigma) \rightarrow C$ be the inclusion map and $\text{en}_\sigma = \text{en}_{\text{supp}_C^P(\sigma)}$. For $a = \|\text{supp}_C^P(\sigma)\|$, the support condition implies the existence of $\sigma_0 \in P(a)$ satisfying $\sigma = P(\epsilon_\sigma \circ \text{en}_\sigma)(\sigma_0)$. By the naturality of ι and supp^Q , we have

$$\begin{aligned} \text{supp}_C^Q(\iota_C(\sigma)) &= \text{supp}_C^Q\left(\iota_C \circ P(\epsilon_\sigma \circ \text{en}_\sigma)(\sigma_0)\right) = \text{supp}_D^Q\left(Q(\epsilon_\sigma \circ \text{en}_\sigma) \circ \iota_a(\sigma_0)\right) \\ &= [\epsilon_\sigma \circ \text{en}_\sigma]^{<\omega}\left(\text{supp}_a^Q(\iota_a(\sigma_0))\right) \subseteq \text{ran } \epsilon_\sigma = \text{supp}_D^P(\sigma). \end{aligned}$$

For the remaining inclusion, suppose that the inclusion fails and there is $x \in \text{supp}_C^P(\sigma) \setminus \text{supp}_C^Q(\iota_C(\sigma))$. Fix $u \in a$ such that $x = \text{en}_\sigma(u)$, then we have $u \in \text{supp}_a^P(\sigma_0) \setminus$

$\text{supp}_a^Q(\iota_a(\sigma_0))$ since

$$\begin{aligned} \text{en}_\sigma(u) &= (\epsilon_\sigma \circ \text{en}_\sigma)(u) \in \text{supp}_C^Q(\iota_C(\sigma))[\epsilon_\sigma \circ \text{en}_\sigma]^{<\omega}[\text{supp}_a^G(\iota_a(\sigma))] \\ &\iff u \in \text{supp}_a^G(\iota_a(\sigma)). \end{aligned}$$

Now let us consider a new semidendrogram b obtained from a by ‘copying’ u and its successors; More precisely, consider a semidendrogram $b \ni a$ with additional nodes v^+ for $v \in a$ extending u , and let us define

1. If w is an immediate predecessor of u , we take $b \models w \multimap u^+$. For two nodes $x, y \in a$ extending u such that $u \models x \multimap y$, we define $b \models x^+ \multimap y^+$.
2. If $x, y \in a$ satisfies $a \models x \leq u \leq y$, we define $b \models x \leq u \leq u^+ \leq y$.
3. $\pi^b(v^+) = \pi^a(v)$ and $\wp_{v^+}^b = \wp_v^a$ for every $v \in a$ extending u .

Now let $f_0: a \rightarrow b$ be the inclusion map, and

$$f_1(x) = \begin{cases} x^+, & \text{if } x \text{ extends } u, \\ x, & \text{otherwise.} \end{cases}$$

Then, on the one hand, we have

- $u = f_0(u) \in [f_0]^{<\omega}[\text{supp}_a^P(\sigma_0)] = \text{supp}_b^P(P(f_0)(\sigma_0))$, and
- $u \notin \text{ran } f_1 \supseteq [f_1]^{<\omega}[\text{supp}_a^P(\sigma_0)] = \text{supp}_b^P(P(f_1)(\sigma_0))$.

Hence $P(f_0)(\sigma_0) \neq P(f_1)(\sigma_0)$. By the injectivity of ι_b , we have

$$Q(f_0)(\iota_a(\sigma_0)) = \iota_b \circ P(f_0)(\sigma_0) \neq \iota_b \circ P(f_1)(\sigma_0) = Q(f_1)(\iota_a(\sigma_0)). \quad (4.6)$$

On the other hand, the support condition also implies the existence of $\tau_0 \in Q(\|\text{supp}_a^Q(\iota_a(\sigma_0))\|)$ satisfying

$$\iota_a(\sigma_0) = Q(\epsilon_{\iota_a(\sigma_0)} \circ \text{en}_{\iota_a(\sigma_0)})(\tau_0)$$

for the inclusion map $\epsilon_{\iota_a(\sigma_0)} : \text{supp}_a^Q(\iota_n(\sigma_a)) \rightarrow a$ and $\text{en}_{\iota_a(\sigma_0)} = \text{en}_{\text{supp}_a^Q(\iota_a(\sigma_0))}$. Since $u \notin \text{ran } \epsilon_{\iota_a(\sigma_0)} = \text{supp}_a^Q(\iota_a(\sigma_0))$, we have

$$f_0 \circ \epsilon_{\iota_a(\sigma)} \circ \text{en}_{\iota_a(\sigma)} = f_1 \circ \epsilon_{\iota_a(\sigma)} \circ \text{en}_{\iota_a(\sigma)}.$$

It implies

$$\begin{aligned} Q(f_0)(\iota_a(\sigma_0)) &= Q\left(f_0 \circ \epsilon_{\iota_a(\sigma_0)} \circ \text{en}_{\iota_a(\sigma_0)}\right)(\tau_0) \\ &= Q\left(f_1 \circ \epsilon_{\iota_a(\sigma_0)} \circ \text{en}_{\iota_a(\sigma_0)}\right)(\tau_0) = Q(f_1)(\iota_a(\sigma_0)), \end{aligned}$$

contradicting (4.6). □

Like the semidilator case, we want to show that the category of 2-semiptykes and that of F-2-semiptykes are equivalent. The following set will have a role of the set of terms in the denotation system we will construct from an F-2-semiptyx:

Definition 4.4.6. The *trace* of an F-2-semiptyx P is given by

$$\text{Tr}(P) = \{(a, \sigma) \mid a \text{ is a natural semidendrogram} \wedge \sigma \in P(a) \wedge \text{supp}_a(\sigma) = a\}.$$

Proposition 4.4.7. For an F-2-semiptyx P , if we define $\mathfrak{f}(P)$ by $\text{field}(\mathfrak{f}(P)) = \text{Tr}(P)$, $\text{arity}^{\mathfrak{f}(P)}(a, t) = a$, and for a 1-arity diagram

$$\mathbb{D} = \begin{array}{ccc} a_{\cap} & \dashrightarrow & a_1 \\ \downarrow & & \downarrow e_1 \\ a_0 & \xrightarrow{e_0} & a_{\cup} \end{array} \quad (4.7)$$

and $(a_0, t_0), (a_1, t_1) \in \text{Tr}(P)$, let us define

$$(a_0, t_0) <_{\mathbb{D}} (a_1, t_1) \iff P(a_{\cup}) \models P(e_0)(t_0) < P(e_1)(t_1). \quad (4.8)$$

Then $\mathfrak{f}(P)$ is a 2-semiptyx.

Proof. It is easy to check that the comparison rule of $\mathfrak{f}(F)$ satisfies asymmetry and linearity. For transitivity, suppose that we have a 1-IU diagram $\mathbb{D}$ for three objects and $t_0, t_1, t_2 \in \text{field}(\mathfrak{f}(F))$ such that $(a_i, t_i) \in \text{Tr}(F)$ for $i = 0, 1, 2$ and

$$\mathfrak{f}(F) \models t_0 <_{\mathbb{D}[0,1]} t_1 <_{\mathbb{D}[1,2]} t_2.$$

Now let $\mathbb{D}(i) = a_i$ and $\mathbb{D}(i, j) = e_{ij}$ for $i \leq j \in \mathbb{L}_3$ be morphisms of $\mathbb{D}$. Then we have

$$P(a_{0\vee 1}) \models P(e_{0,0\vee 1})(t_0) < P(e_{1,0\vee 1})(t_1) \text{ and } P(a_{1\vee 2}) \models P(e_{1,1\vee 2})(t_1) < P(e_{2,1\vee 2})(t_2).$$

By applying $P(e_{0\vee 1,0\vee 1\vee 2})$ to the left side, we have

$$P(a_{0\vee 1\vee 2}) \models P(e_{0,0\vee 1\vee 2})(t_0) < P(e_{1,0\vee 1\vee 2})(t_1).$$

Similarly, we can see

$$P(a_{0\vee 1\vee 2}) \models P(e_{1,0\vee 1\vee 2})(t_1) < P(e_{2,0\vee 1\vee 2})(t_2).$$

Thus we have

$$P(a_{0\vee 1\vee 2}) \models P(e_{0,0\vee 1\vee 2})(t_0) < P(e_{2,0\vee 1\vee 2})(t_2).$$

Then by subtracting $P(e_{0\vee 2,0\vee 1\vee 2})$, we get

$$P(a_{0\vee 2}) \models P(e_{0,0\vee 2})(t_0) < P(e_{2,0\vee 2})(t_2),$$

which is equivalent to $P \models t_0 <_{\mathbb{D}[0,2]} t_2$. $\square$

We can turn $\mathfrak{f}$ into a functor $\mathbf{SPtyx}_{\mathbb{F}}^2 \rightarrow \mathbf{SPtyx}^2$ by defining $\mathfrak{f}(\iota)$ for a natural transformation $\iota: P \rightarrow Q$ by

$$\mathfrak{f}(\iota)(a, t) = (a, \iota_a(t)).$$

Lemma 4.4.8. $\mathfrak{f}(\iota)$ is a 2-semiptyx embedding from $\mathfrak{f}(P)$ to $\mathfrak{f}(Q)$.

Proof. $\mathfrak{f}(\iota)$ preserves an arity of a term. To see the comparison rule preservation, suppose we are given a 1-arity diagram $\mathbb{D}$ of the form (4.7), and assume that we are given $\mathfrak{f}(P) \models (a_0, t_0) <_{\mathbb{D}} (a_1, t_1)$, or equivalently,

$$P(a_{\cup}) \models P(e_0)(t_0) < P(e_1)(t_1).$$

Then let us apply $\iota_{a_{\cup}}: P(a_{\cup}) \rightarrow Q(a_{\cup})$. Since ι is a natural transformation, we have $\iota_{a_{\cup}} \circ P(e_0) = Q(e_0) \circ \iota_{a_0}$ and $\iota_{a_{\cup}} \circ P(e_1) = Q(e_1) \circ \iota_{a_1}$. Hence

$$Q(a_{\cup}) \models Q(e_0)(\iota_{a_0}(t_0)) < Q(e_1)(\iota_{a_1}(t_1)),$$

which is the definition of $\mathfrak{f}(Q) \models \mathfrak{f}(\iota)(a_0, t_0) <_{\mathbb{D}} \mathfrak{f}(\iota)(a_1, t_1)$. $\square$

Conversely, we can construct a functor $\mathfrak{a}(P)$ from an F-2-semiptyx in a natural way:

Definition 4.4.9. Let P be a 2-semiptyx. Let us define $\mathfrak{a}(P)(C) = P(C)$ for a semidendrogram C . For a semidendrogram morphism $f: C \rightarrow C'$, let us define $\mathfrak{a}(P)(f): \mathfrak{a}(P)(C) \rightarrow \mathfrak{a}(P)(C')$ by

$$\mathfrak{a}(P)(f)(t(a)) = t(f[a]).$$

for $t \in P$ and finite $a \subseteq C$ such that $\text{arity } t = \|a\|$.

Then $\mathfrak{a}(P): \text{SDil} \rightarrow \text{LO}$ is a functor. Also, for a 2-semiptyx embedding $\iota: P \rightarrow Q$, let us define the natural transformation $\mathfrak{a}(\iota): \mathfrak{a}(P) \rightarrow \mathfrak{a}(Q)$ by

$$\mathfrak{a}(\iota)_C(t(a)) = \iota(t)(a).$$

for $t \in \text{field}(P)$ and $a \in [D]^{\text{arity } t}$. To turn $\mathfrak{a}(P)$ an F-2-semiptyx, we need a support transformation: Let us define $\text{supp}^{\mathfrak{a}(P)}: \mathfrak{a}(P) \rightarrow [\cdot]^{<\omega}$ by

$$\text{supp}_D^{\mathfrak{a}(P)}(t(a)) = a.$$

Lemma 4.4.10. $\text{supp}^{\mathfrak{a}(P)}$ satisfies the support condition, so $\mathfrak{a}(P)$ is an F-2-semiptyx.

Proof. Let C, C' be semidendrograms, and $f: C \rightarrow C'$ be a semidendrogram embedding. Suppose that $t \in P$ and $a \subseteq C$, $\|a\| = \text{arity } t$, and

$$\text{supp}_C^{\mathfrak{a}(E)}(t(a)) = a \subseteq \text{ran } f.$$

Since f is one-to-one, there is $b \in [D]^{<\omega}$ such that $a = f[b]$, so $t(a) = \mathfrak{a}(P)(f)(t(b))$ by definition. □

As we did in [Section 1.4](#), we constructed functors $\mathfrak{f}: \text{SPtyx}_{\mathbb{F}}^2 \rightarrow \text{SPtyx}^2$ and $\mathfrak{a}: \text{SPtyx}^2 \rightarrow \text{Ptyx}_{\mathbb{F}}^2$. We prove they are category equivalences:

Theorem 4.4.11. *There is a natural isomorphism $\text{eaf}: \mathfrak{a} \circ \mathfrak{f} \rightarrow \text{Id}_{\text{SPtyx}_{\mathbb{F}}^2}$.*

Proof. Let us describe $\mathfrak{a}(\mathfrak{f}(P))$ for an F-2-semiptyx P : $\mathfrak{a}(\mathfrak{f}(P))$ is a functor from SDil to LO, which maps a semidendrogram C to a linear order of field

$$\mathfrak{a}(\mathfrak{f}(P))(C) = \{(a, t)(c) \mid (a, t) \in \text{Tr}(F) \wedge a \cong c \subseteq C\}.$$

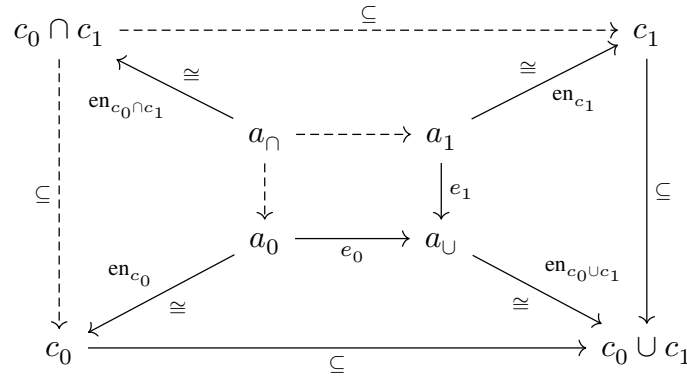
The comparison rule over $\mathfrak{a}(\mathfrak{f}(F))(X)$ is as follows: For $(a_0, t_0), (a_1, t_1) \in \text{Tr}(F)$ and $c_0, c_1 \subseteq C$,

$$\begin{aligned} \mathfrak{a}(\mathfrak{f}(P))(C) \models (a_0, t_0)(c_0) < (a_1, t_1)(c_1) \\ \iff \text{For } \mathbb{D} = \text{Diag}_C(c_0, c_1), \mathfrak{f}(P) \models (a_0, t_0) <_{\mathbb{D}} (a_1, t_1). \end{aligned}$$

and if $\mathbb{D}$ takes the form (4.7),

$$\mathfrak{f}(P) \models (a_0, t_0) <_{\mathbb{D}} (a_1, t_1) \iff P(a_{\cup}) \models P(e_0)(t_0) < P(e_1)(t_1).$$

Note that $\mathbb{D} = \text{Diag}_C(c_0, c_1)$ implies $a_{\cap} = \|a_0 \cap a_1\|$, $a_{\cup} = \|a_0 \cup a_1\|$ and the following diagram commutes:



Hence applying $P(\text{en}_{c_0 \cup c_1})$ to $P(n_{\cup}) \models P(e_0)(t_0) < P(e_1)(t_1)$ gives

$$\mathfrak{f}(P) \models (a_0, t_0) <_{\mathbb{D}} (a_1, t_1) \iff P(c_0 \cup c_1) \models P(\text{en}_{c_0})(t_0) < P(\text{en}_{c_1})(t_1).$$

Hence

$$\mathfrak{a}(\mathfrak{f}(P))(C) \models (a_0, t_0)(c_0) < (a_1, t_1)(c_1) \iff P(c_0 \cup c_1) \models P(\text{en}_{c_0})(t_0) < P(\text{en}_{c_1})(t_1).$$

Now let us define a natural transformation $\mathbf{eaf}_P: (\mathfrak{a} \circ \mathfrak{f})(P) \rightarrow P$ as follows: For each semidendrogram C , $\mathbf{eaf}_{P,C}: (\mathfrak{a} \circ \mathfrak{f})(P)(C) \rightarrow P(C)$ is the map

$$\mathbf{eaf}_{P,C}((a, t)(c)) = P(\mathbf{en}_c)(t).$$

Then $\mathbf{eaf}_{P,C}$ is an isomorphism: It is order-preserving since

$$\begin{aligned} (\mathfrak{a} \circ \mathfrak{f})(P)(C) &\models (a_0, t_0)(c_0) < (a_1, t_1)(c_1) \\ \iff P(c_0 \cup c_1) &\models P(\mathbf{en}_{c_0})(t_0) < P(\mathbf{en}_{c_1})(t_1) \\ \iff P(C) &\models P(\mathbf{en}_{c_0})(t_0) < P(\mathbf{en}_{c_1})(t_1). \end{aligned}$$

$\mathbf{eaf}_{P,C}$ is onto since for every $u \in P(C)$, if we let $c = \text{supp}_C^P(u)$ and $t = P(\mathbf{en}_c^{-1})(u)$, then $(\|c\|, t) \in \text{Tr}(P)$ and $\mathbf{eaf}_{P,C}((\|c\|, t)(c)) = u$.

To see $\mathbf{eaf}_P$ is a natural transformation from $\mathfrak{a}(\mathfrak{f}(P))$ to P , let us observe that what $\mathfrak{a}(\mathfrak{f}(P))(f)$ should be for a semidendrogram morphism $f: C \rightarrow C'$: By definition of $\mathfrak{a}$, we have

$$\mathfrak{a}(\mathfrak{f}(P))(f)((a, t)(c)) = (a, t)(f[c]).$$

Hence for $(a, t)(c) \in \mathfrak{a}(\mathfrak{f}(P))(C)$,

$$\begin{aligned} P(f) \circ \mathbf{eaf}_{P,C}((a, t)(c)) &= P(f)(P(\mathbf{en}_c)(t)) = P(f \circ \mathbf{en}_c)(t) \\ &= P(\mathbf{en}_{f[c]})(t) = \mathbf{eaf}_{P,E}((a, t)(f[c])) \\ &= (\mathbf{eaf}_{P,C'} \circ \mathfrak{a}(\mathfrak{f}(P))(f))((a, t)(c)), \end{aligned}$$

proving the naturality of $\mathbf{eaf}_P$. However, we also need that $\mathbf{eaf}_{P,C}$ is natural also in P . To see this, we claim that the following diagram commutes for every natural transformation $\iota: P \rightarrow Q$ and a semidilator D :

$$\begin{array}{ccc} \mathfrak{a}(\mathfrak{f}(P))(C) & \xrightarrow{\mathfrak{a}(\mathfrak{f}(\iota))_C} & \mathfrak{a}(\mathfrak{f}(Q))(C) \\ \mathbf{eaf}_{P,C} \downarrow & & \downarrow \mathbf{eaf}_{Q,C} \\ P(C) & \xrightarrow{\iota_C} & Q(C) \end{array}$$

For each $(a, t)(c) \in \mathfrak{f}(P)(C)$, note that $a = \|d\|$, so

$$\begin{aligned} \mathbf{eaf}_{Q,C}(\mathfrak{a}(\mathfrak{f}(\iota))_C((a, t)(c))) &= \mathbf{eaf}_{Q,C}(\mathfrak{f}(\iota)(a, t)(c)) = \mathbf{eaf}_{Q,D}((a, \iota_{\|c\|}(t))(c)) \\ &= Q(\mathbf{en}_c)(\iota_{\|c\|}(t)) = \iota_C(P(\mathbf{en}_d)(t)) \\ &= \iota_C(\mathbf{eaf}_{P,C}((a, t)(c))), \end{aligned}$$

here the linebroke equality follows from $Q(\mathbf{en}_c) \circ \iota_{\|a\|} = \iota_C \circ P(\mathbf{en}_c)$ since $\iota: P \rightarrow Q$ is a natural transformation. It finishes the proof. $\square$

Theorem 4.4.12. *There is a natural isomorphism $\mathbf{efa}: \mathbf{Id}_{\mathbf{SPtyx}^2} \rightarrow \mathfrak{f} \circ \mathfrak{a}$.*

Proof. Let us observe the structure of $\mathfrak{f}(\mathfrak{a}(P))$ for a 2-semiptyx P . Its field is

$$\begin{aligned} \mathrm{Tr}(\mathfrak{a}(P)) &= \{(a, t) \mid a \in \mathbf{SNat}^2 \wedge t \in P(a) \wedge \mathrm{supp}_a^{\mathfrak{a}(P)}(t) = a\} \\ &= \{(a, t(a)) \mid a \in \mathbf{SNat}^2 \wedge t \in \mathrm{field}(P) \wedge \mathrm{arity}^P(t) = a\} \end{aligned}$$

and $\mathrm{arity}^{\mathfrak{f}(\mathfrak{a}(P))}(a, t(\mathrm{Dec}(a))) = a$. For a 1-arity diagram $\mathbb{D}$ of the form (1.5),

$$\begin{aligned} \mathfrak{f}(\mathfrak{a}(P)) \models (a_0, t_0(n_0)) <_{\mathbb{D}} (a_1, t_1(n_1)) &\iff P(a_{\cup}) \models P(e_0)(t_0(a_0)) < P(e_1)(t_1(a_1)) \\ &\iff P(a_{\cup}) \models t_0(e_0[a_0]) < t_1(e_1[a_1]), \end{aligned}$$

and observe that $\mathrm{Diag}_{a_{\cup}}(e_0[a_0], e_1[a_1]) = \mathbb{D}$, so

$$P(a_{\cup}) \models t_0(e_0[a_0]) < t_1(e_1[a_1]) \iff P \models t_0 <_{\mathbb{D}} t_1.$$

That is, $\mathfrak{f}(\mathfrak{a}(P)) \models (a_0, t_0(a_0)) <_{\mathbb{D}} (a_1, t_1(a_1)) \iff P \models t_0 <_{\mathbb{D}} t_1$. Hence if we let $\mathbf{efa}_P: P \rightarrow \mathfrak{f}(\mathfrak{a}(P))$ by $\mathbf{efa}_P(t) = (\mathrm{arity} t, t(\mathrm{arity} t))$, then $\mathbf{efa}_P$ is a 2-semiptyx isomorphism. The naturality of $\mathbf{efa}$ follows from an easy computation. $\square$

Hence we have

Corollary 4.4.13. *$\mathbf{SPtyx}^2$ and $\mathbf{SPtyx}_{\mathbb{F}}^2$ are category equivalent.* $\square$

Thus we can identify the two categories SPtyx^2 and SPtyx_F . Also, let us observe that, like semidilator cases, the construction of $\mathfrak{f}(P)$ does not require the whole P . In fact, we can construct $\mathfrak{f}(P)$ from the restriction of P to the category SNat^2 of natural semidilators, a full subcategory of the category of 2-semiptykes. Hence we can construct a 2-semiptyx not only from a whole F-2-semiptyx but also from its restriction to the SNat^2 .

Definition 4.4.14. An *F-coded 2-semiptyx* is a pair (P, supp) , where $P: \text{SNat}^2 \rightarrow \text{LO}$ is a functor and supp is a natural transformation with the support condition (1.4). If P is positive and monotone, we say P is an *F-coded 2-preptyx*.

The previous proofs give the following corollary:

Corollary 4.4.15. *If P is an F-coded 2-semiptyx, then we can construct a 2-semiptyx $\mathfrak{f}(P)$. If P is an F-2-semiptyx, then $\mathfrak{f}(P) = \mathfrak{f}(P \upharpoonright \text{SNat}^2)$ and $P \cong \mathfrak{a}(\mathfrak{f}(P \upharpoonright \text{SNat}^2))$. $\square$*

If P is positive, then $P(a) \cong 0$ for every $a \in \text{SNat}^2 \setminus \text{Nat}^2$. Hence we also have

Corollary 4.4.16. *If P is an F-2-preptyx, then $\mathfrak{f}(P) = \mathfrak{f}(P \upharpoonright \text{Nat}^2)$ and $P \cong \mathfrak{a}(\mathfrak{f}(P \upharpoonright \text{Nat}^2))$. That is, we can reconstruct the full 2-preptyx from its restriction to the category of natural dilators. $\square$*

Like dilators, we can also discuss whether a given F-coded 2-semiptyx is recursive. We also have

Proposition 4.4.17. *Suppose that P is an F-coded 2-semiptyx that is coded by a (primitive) recursive subset of $\mathbb{N}$. Then $\mathfrak{f}(P)$ is (primitive) recursive. $\square$*

4.5 Prime 2-ptykes

In this section, we classify prime 2-ptykes, a 2-ptyx with a single term.

Definition 4.5.1. A 2-ptyx P is *prime* if it has a unique term. The *degree* of P is the arity of the unique P -term.

Like prime dilators, we find a ‘priority permutation’ for a prime 2-ptyx, which is no longer a permutation. A priority permutation assigns which component must be compared first. The 2-ptyx analogue of the priority assignment assigns which node over a dendrogram expression must be compared first. In the proof of [Theorem 2.2.4](#), we prepared subsidiary sequences c_i over $2n$, where n is the arity of the unique term in the prime dilator. Before characterizing prime 2-ptykes, we define a predendrogram analogue of $n \mapsto 2n$ and c_i :

Definition 4.5.2. Let C be a predendrogram. We define the *duplication* $\text{dup}(C)$ of C as a predendrogram defined as follows: The field of $\text{dup}(C)$ is obtained from that of C by adding v^{+u} for each $u \in C$ and $v \in C$ extending u (including u itself.) Informally, the plus tree $\{v^{+u} \mid v \text{ extends } u\}$ with root u will form a duplicated copy of a subtree $\{v \mid v \text{ extends } u\}$ of root u in the dendrogram $\text{dup}(C)$, and we put the plus tree of root u at the right side of the subtree of root u .

We define relations over $\text{dup}(C)$ as follows:

1. C is a sub-predendrogram of $\text{dup}(C)$, so $\text{dup}(C)$ respects every relation over C .
2. If $C \models u \multimap v$, then $\text{dup}(C) \models u \multimap v^{+v}$.
3. For $u, v, w \in C$ such that both v and w extends u and $C \models v \multimap w$, $\text{dup}(C) \models v^{+u} \multimap w^{+u}$.
4. If $C \models u \leq v$, then $\text{dup}(C) \models u \leq u^{+u} \leq v \leq v^{+v}$.
5. For $u, v, w \in C$ such that both v and w extends u and $C \models v \leq w$, $\text{dup}(C) \models v^{+u} \leq w^{+u}$.
6. If v extends u , $\pi^{\text{dup}(C)}(v^{+u}) = \pi^C(v)$.

We may understand the construction of $\text{dup}(C)$ as follows: First, let us color the current term part black. In level m , copy each black-colored $u \in C$ and its successors and put the copy just right of the original term. We color the copied term parts white.

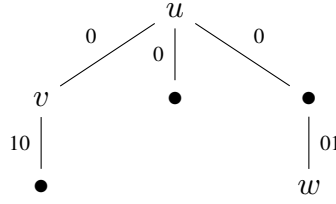


Figure 4.2: Square dendrogram

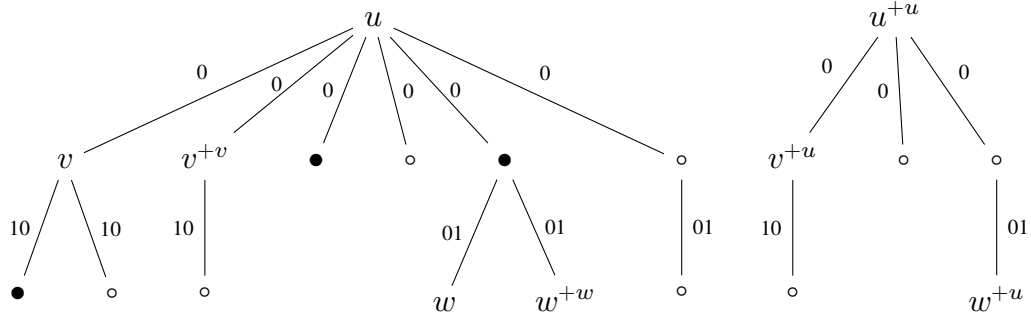


Figure 4.3: Duplicated square dendrogram

For $u \in C$, let $u^{+u} \in \text{dup}(C)$ be the copy of u placed just right of u . Also, we denote successors of u^{+u} corresponding to $v \supseteq u$ by v^{+u} .

Example 4.5.3. Let us recall the natural dendrogram d isomorphic to the dilator $X \mapsto X \times X$ presented in Figure 3.1. We reintroduce its figure as in Figure 4.2 for convenience, with a different labeling on nodes for a later purpose. Figure 4.3 represents the duplication of the square dendrogram.

Definition 4.5.4. Let c be a natural predendrogram and $u \in c$. We define $\mathbf{c}_{c,u} \subseteq \text{dup}(c)$ as a subdilator isomorphic to c given by

$$\mathbf{c}_{c,u} = \{w^{+u} \mid w \in c \wedge u \multimap^* w\} \cup \{w \mid w \in c \wedge w \not\multimap^* u\}.$$

Example 4.5.5. Let c be the dendrogram representing $X \mapsto X \times X$ provided in Figure 4.2. Recalling its duplication in Figure 4.3, we can represent $\mathbf{c}_{c,u}$, $\mathbf{c}_{c,v}$, $\mathbf{c}_{c,w}$ as in Figure 4.4.

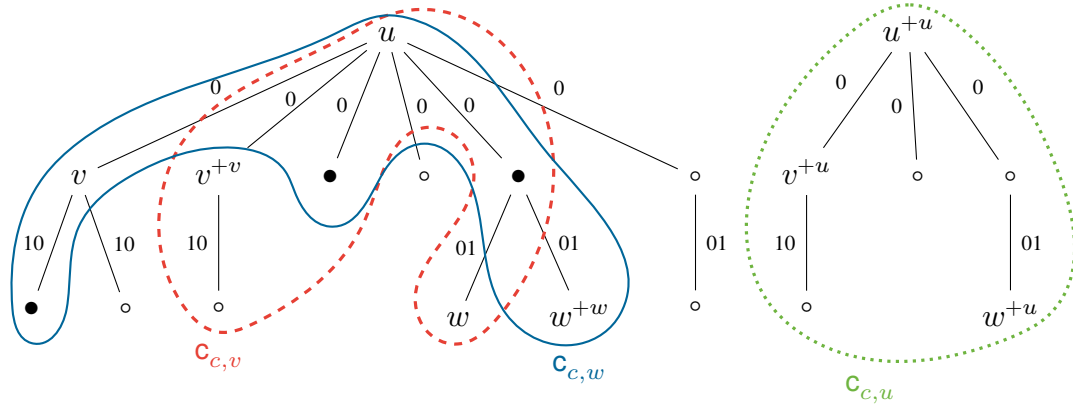


Figure 4.4: Examples of subdendrograms of Figure 4.3

Let us recall the details from Section 2.2: We tracked the priority order (equivalently, the priority permutation) of a given term t of arity m by comparing $t(c_0), \dots, t(c_{m-1})$. The priority permutation can be arbitrary. It corresponds to the fact that $\text{Diag}_{2m}(c_i, c_j)$ is not monotone if $i \neq j$. (See the proof of Theorem 2.2.4 for the definition of c_i .) However, unlike the previous case, $\text{Diag}_{\text{dup}(c)}(c_{c,u}, c_{c,v})$ can be monotone: We can see that $\text{Diag}_{\text{dup}(c)}(c_{c,v}, c_{c,u})$ in Figure 4.4 is monotone. In general, $\text{Diag}_{\text{dup}(c)}(c_{c,u}, c_{c,v})$ is monotone when u is extended from v . It will come as a constraint on defining a prime 2-ptyx from a permutation since if $P = \{t\}$ is a prime 2-ptyx of degree c and u is extended from v , then we must have

$$P(c) \models t(c_{c,u}) < t(c_{c,v}).$$

It shows that ‘priority permutations’ for prime 2-ptykes cannot be as arbitrary as priority permutations for prime dilators. However, the following lemma says $u \circ^* v$ is the only case when the diagram becomes monotone:

Lemma 4.5.6. *Let c be a natural dendrogram and $u, v \in c$. Then $\text{Diag}_{\text{dup}(c)}(c_{c,u}, c_{c,v})$ is monotone if and only if $u \circ^* v$.*

Proof. Suppose that $u \circ^* v$. Then we have $\text{en}_{c_{c,u}}(w) = \text{en}_{c_{c,v}}(w)$ if $w \not\circ^* v$ or $w \circ u$.

If $w \circ^* v$ but $w \not\circ^* u$, then $\text{en}_{c,u}(w) = w \leq_{\text{KB}} \text{en}_{c,v}(w)$. Hence for every $w \in c$, we have $\text{en}_{c,u}(w) \leq_{\text{KB}} \text{en}_{c,v}(w)$, proving the monotonicity of the diagram. Conversely, if $u \not\circ^* v$, then we have $\text{en}_{c,u}(u) = u^{+u} > u = \text{en}_{c,v}(u)$, violating the monotonicity. $\square$

We expect the following notion to characterize a prime 2-ptyx of given arity:

Definition 4.5.7. A finite (pre)dendrogram c is a *priority (pre)dendrogram* if the field of c is equal to its cardinality $|c|$, and for every $i, j < |c|$, $c \models i \multimap j$ implies $i < j$ as natural numbers.

As in priority orders, nodes with lower indices have higher priority. It is easy to see that if c is a priority dendrogram and $c \models i \multimap^* j$, then $i \leq j$. It turns out that every priority (pre)dendrogram c induces a prime 2-ptyx of degree $\|c\|$:

Example 4.5.8. Let c be a priority dendrogram. Let us define Prime_c as follows: For a given semidendrogram D , let us define $\text{Prime}_c(D)$ be the set of every sub-predendrogram of D isomorphic to c , i.e.,

$$\text{Prime}_c(D) = \{a \subseteq D \mid a \cong c\}.$$

We equip $\text{Prime}_c(D)$ with the order $<_c$ given as follows. We first define, for two isomorphic finite predendrograms a and c ,

$$\text{en}_{c,a} = \text{en}_a \circ \text{en}_c^{-1} : c \rightarrow a. \quad (4.9)$$

Then for two $a, b \in \text{Prime}_c(D)$ isomorphic to c , $a <_c b$ if and only if $i < |c|$ is the least number such that $\text{en}_{c,a}(i) \neq \text{en}_{c,b}(i)$, then $D \models \text{en}_{c,a}(i) \leq \text{en}_{c,b}(i)$. Note that for the least i with $\text{en}_{c,a}(i) \neq \text{en}_{c,b}(i)$, either both $\text{en}_{c,a}(i)$ or $\text{en}_{c,b}(i)$ have length 0, or have the same immediate predecessor: The immediate predecessor of $f_a(i)$, if exists, is $\text{en}_{c,a}(j)$ for some $j < i$, so $\text{en}_{c,a}(j) = \text{en}_{c,b}(j)$. From $c \models j \multimap i$, we also have $D \models \text{en}_{c,b}(j) \multimap \text{en}_{c,b}(i)$. The support transformation is defined trivially, i.e., $\text{supp}_D^{\text{Prime}_c}(a) = a$. For a

semidendrogram embedding $f: C \rightarrow D$, we define $\text{Prime}_c(f)(a) = f[a]$. We can easily check the support condition of Prime_c .

We can see that Prime_c is a 2-ptyx. In fact, we can see that if D is a locally well-founded predendrogram, then $\text{Prime}_c(D)$ is well-founded. Suppose the contradiction that $\text{Prime}_c(D)$ is ill-founded, so we have $a_m \subseteq D$ ($m \in \mathbb{N}$) such that $a_0 >_c a_1 >_c a_2 >_c \dots$. Let $h_m = \text{en}_{c, a_m} : c \rightarrow a_m$ be the unique isomorphism from c to a_m for each m . We claim by induction on $i < |c|$ that $\langle h_m(i) \mid m \in \mathbb{N} \rangle$ is eventually constant: Suppose that either i is a root in c , or $c \models j \multimap i$ for some $j < i$, and we inductively have that $\langle h_m(j) \mid m \in \mathbb{N} \rangle$ is eventually constant. In the latter case, we may assume that $\langle h_m(j) \mid m \in \mathbb{N} \rangle$ is constant. In either case, we have

$$D \models h_0(i) \geq h_1(i) \geq h_2(i) \geq \dots,$$

and if $\langle h_m(i) \mid m \in \mathbb{N} \rangle$ is not eventually constant, then it contradicts the local well-foundedness of D .

From the remaining part of this section, we prove the following theorem:

Theorem 4.5.9. *Let P be a prime 2-ptyx of degree d . Then there is a priority (pre)dendrogram c isomorphic to d such that $P \cong \text{Prime}_c$.*

Proof. Suppose that t is the unique P -term of arity d . We define the priority predendrogram c by relabeling d as follows: First, let us enumerate d as $\{u_i \mid i < |d|\}$ in a way that

$$P(\text{dup}(d)) \models t(\mathbf{c}_{d, u_0}) > t(\mathbf{c}_{d, u_1}) > t(\mathbf{c}_{d, u_2}) > \dots.$$

Then define c in a way that the map $u_i \mapsto i$ is a predendrogram isomorphism from d to c . Lemma 4.5.6 implies that c is a priority (pre)dendrogram.

We claim that for a semidendrogram D and $a, b \subseteq D$ isomorphic to d ,

$$P(D) \models t(a) < t(b) \iff \exists i < |c| [\forall j < i [\text{en}_{c, a}(j) = \text{en}_{c, b}(j)] \wedge \text{en}_{c, a}(i) < \text{en}_{c, b}(i)]. \quad (4.10)$$

Note that if $a \subseteq D$ is isomorphic to d , then $a \subseteq \text{pos}(D)$. Hence, we may assume that D is a predendrogram. In the remaining part of the proof, we assume D is a predendrogram.

By linearity of P , it suffices to show the backward implication of (4.10). For a natural predendrogram d , let $\ell(d)$ be the height of d , or equivalently, the maximum length of nodes of d plus 1. For each $k < \ell(d)$ and $a, b \subseteq D$ isomorphic to d , let $N_{a,b}(k)$ be the number of $u \in d$ of length k such that $\text{en}_a(u) \succ_{\text{KB}} \text{en}_b(u)$. Then define

$$N_{a,b} = |d|^{\ell(d)-1} \cdot N_{a,b}(0) + |d|^{\ell(d)-2} \cdot N_{a,b}(1) + \cdots + N_{a,b}(\ell(d) - 1). \quad (4.11)$$

We may regard $N_{a,b}$ as coding the tuple $\langle N_{a,b}(k) \mid k < \ell(d) \rangle$ in lexicographic order, with the number of inversions at smaller lengths taking precedence.

We prove the backward implication of (4.10) by induction on $N_{a,b}$. If $N_{a,b} = 0$, then $\text{en}_a(u) \leq_{\text{KB}} \text{en}_b(u)$ for every $u \in d$, so $P(D) \models t(a) \leq t(b)$ since every 2-ptyx is monotone. Since the right-hand side of (4.10) implies $a \neq b$, we have $P(D) \models t(a) < t(b)$.

Now suppose that $N_{a,b} > 0$, and let i witness the right-hand side of (4.10). Choose $l < |c|$ such that $\text{en}_{c,a}(l) \succ_{\text{KB}} \text{en}_{c,b}(l)$, $\text{lh}^c(l)$ is minimal among all nodes with this property, and, among all such nodes of length $\text{lh}^c(l)$, l is the leftmost one.

We first make a few observations about l . The following two imply $i < l$:

1. $\text{en}_{c,a}(j) = \text{en}_{c,b}(j)$ for every $j < i$.
2. $\text{en}_{c,a}(i) < \text{en}_{c,b}(i)$.

Moreover, for every immediate predecessor l' of l in c , we have $\text{en}_{c,a}(l') = \text{en}_{c,b}(l')$, otherwise, $\text{en}_{c,a}(l)$ and $\text{en}_{c,b}(l)$ are $<$ -incomparable: $l \rightarrow^* i$ would imply $l < i$, and $i \rightarrow^* l$ would imply $\text{en}_{c,a}(i) < \text{en}_{c,b}(i)$, so $\text{en}_{c,a}(l)$ and $\text{en}_{c,b}(l)$ are $<$ -incomparable.

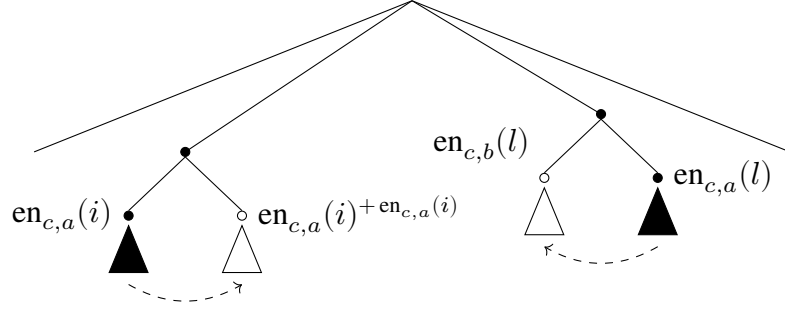


Figure 4.5: The construction of $\bar{a}$ from a . We replace filled nodes and triangles with hollow nodes and triangles to get $\bar{a}$ from a , with all other parts unchanged.

Let us define $\bar{a} \subseteq \text{dup}(D)$ isomorphic to d by specifying $\text{en}_{c,\bar{a}}: c \rightarrow \bar{a}$ as follows:

$$\text{en}_{c,\bar{a}}(j) = \begin{cases} \text{en}_{c,a}(j) + \text{en}_{c,a}(i) & c \models j \circ^* i, \\ \text{en}_{c,b}(j) & c \models j \circ^* l, \\ \text{en}_{c,a}(j) & \text{otherwise.} \end{cases}$$

(See Figure 4.5 for the illustration of $\bar{a}$.)

We claim that $\bar{a}$ is isomorphic to c . Since $\text{en}_{c,a}(l') = \text{en}_{c,b}(l')$ for $c \models l' \rightarrow l$, replacing the subtree of $\text{en}_{c,a}(l)$ with the corresponding subtree of $\text{en}_{c,b}(l)$ preserves the predecessor relation. Furthermore, if $c \models k \prec l \prec m$, then k is not an inversion, by the choice of l . Hence

$$\text{dup}(D) \models \text{en}_{c,a}(k) \leq \text{en}_{c,b}(k) \prec \text{en}_{c,b}(l) \prec \text{en}_{c,a}(l) \prec \text{en}_{c,a}(m).$$

Thus the replacement at l preserves the local comparison order. The replacement at i also preserves the local order, since

$$\text{dup}(D) \models \text{en}_{c,a}(i) \prec \text{en}_{c,a}(i) + \text{en}_{c,a}(i) \prec \text{en}_{c,b}(i).$$

Therefore $\bar{a}$ is isomorphic to c . We claim that

$$P(\text{dup}(D)) \models t(a) < t(\bar{a}) < t(b).$$

First, let us prove $P(\text{dup}(D)) \models t(\bar{a}) < t(b)$. Let $r = \text{lh}^c(l)$. By the minimality of $\text{lh}^c(l)$, $N_{a,b}(k) = 0$ for every $k < r$. The modification of the subtree above i introduces no new inversion, since the duplicated subtree rooted at $\text{en}_{c,a}(i)^{+\text{en}_{c,a}(i)}$ still lies entirely to the left of the corresponding subtree rooted at $\text{en}_{c,b}(i)$. The modification above l replaces the a -subtree by the corresponding b -subtree. Hence $N_{\bar{a},b}(k) = N_{a,b}(k)$ for every $k < r$ and $N_{\bar{a},b}(r) = N_{a,b}(r) - 1$. Therefore $N_{\bar{a},b} < N_{a,b}$.

Moreover, the same i witnesses the right-hand side of (4.10) for $\bar{a}$ and b : all coordinates below i are unchanged and

$$\text{en}_{c,\bar{a}}(i) = \text{en}_{c,a}(i)^{+\text{en}_{c,a}(i)} \leq \text{en}_{c,b}(i).$$

Hence, the induction hypothesis yields $P(\text{dup}(D)) \models t(\bar{a}) < t(b)$, as desired.

It remains to prove $P(\text{dup}(D)) \models t(a) < t(\bar{a})$. Let us consider a predendrogram morphism $g: \mathbf{c}_{c,i} \cup \mathbf{c}_{c,l} \rightarrow a \cup \bar{a}$ given by

- $g(j) = \text{en}_{c,a}(j)$ for $c \models j \circ^* i$.
- $g(j^{+i}) = \text{en}_{c,a}(j)^{+\text{en}_{c,a}(i)}$ for $c \models j \circ^* i$.
- $g(j) = \text{en}_{c,b}(j)$ for $c \models j \circ^* l$.
- $g(j^{+l}) = \text{en}_{c,a}(j)$ for $c \models j \circ^* l$.
- $g(j) = \text{en}_{c,a}(j)$ otherwise.

Then g is an isomorphism, $g[\mathbf{c}_{c,i}] = \bar{a}$, and $g[\mathbf{c}_{c,l}] = a$. Since $i < l$, we have

$$P(\text{dup}(c)) \models t(\mathbf{c}_{c,i}) > t(\mathbf{c}_{c,l}).$$

Applying $P(g)$ yields $P(\text{dup}(D)) \models t(\bar{a}) > t(a)$.

Finally, the inclusion $D \rightarrow \text{dup}(D)$ is a predendrogram embedding, and the induced map $P(D) \rightarrow P(\text{dup}(D))$ is an order embedding. Therefore $P(D) \models t(a) < t(b)$, proving the backward implication of (4.10). $\square$

As we did in [Section 2.2](#), we have the following result:

Theorem 4.5.10. *Let P be a 2-preptyx. For each $t \in \text{field}(P)$ of arity d , we can find a priority (pre)dendrogram $c = \pi^P(t)$ such that for every predendrogram D and $a, b \subseteq D$ isomorphic to $\pi^D(t)$, $P(D) \models t(a) < t(b)$ if and only if*

$$\exists i < |c| [\forall j < i [\text{en}_{c,a}(j) = \text{en}_{c,b}(j)] \wedge \text{en}_{c,a}(i) < \text{en}_{c,b}(i)]. \quad \square$$

4.6 The comparison structure of 2-preptykes

In this section, we further analyze the comparison between distinct terms of a 2-preptyx. By the characterization of prime 2-ptykes in the previous section, each P -term s comes equipped with a priority predendrogram $\pi^P(s)$. We want to associate with two distinct P -terms s, t a number $\mathbf{p}^P(s, t)$ which measures how far the arguments of s and t have to be compared according to the priority predendrograms $\pi^P(s)$ and $\pi^P(t)$.

For predilators, the analogous number can be characterized by asking how many initial arguments have to agree before the comparison between two terms becomes independent of the remaining arguments. For 2-preptykes, this description is no longer sufficient. There are two additional phenomena, both caused by the fact that priority predendrograms contain nodes which are not themselves genuine arguments of a term.

First, a node which is terminal in the arity of one term may correspond to a non-terminal node in the arity of the other. Thus the agreement which is needed to analyze the comparison may fail to be realizable by genuine arguments, even though it can be realized after we temporarily ignore terminality.

Second, there is a stronger obstruction. Even after terminality is ignored, two initial parts of the priority predendrograms need not admit a common realization. Identifying two non-terminal nodes may force their immediate successors to have two different priority orders, which is impossible in a predendrogram. Consequently, at a given stage there

are three possibilities: the comparison may still vary in both directions, it may already have a fixed direction, or the required agreement itself may be impossible.

The following example illustrate the first two phenomena.

Example 4.6.1. Consider the evaluation ptyx $P(D) = D(3)$. Its trace consists of 16 terms. We index them by the finite words without repetition over 0, 1, 2:

$$\emptyset, \quad 0, 1, 2, \quad 01, 02, 12, 10, 20, 21, \quad 012, 021, 102, 120, 201, 210.$$

We denote the term corresponding to a word p by τ_p . The word p records two kinds of information simultaneously. Its letters specify which elements of 3 are substituted into the underlying dilator term, while their order records the priority order in which these arguments are compared. More precisely, if $p = \langle p_0, \dots, p_{n-1} \rangle$, then τ_p has arity $\text{Prime}_{\text{std}(p)}$, where $\text{std}(p)$ is the permutation of n obtained by replacing the entries of p by their relative ranks. Thus, informally, τ_p takes a term t of this arity and evaluates it as $t(p_0, \dots, p_{n-1})$ with the order of the letters of p simultaneously recording the priority of the corresponding arguments.

For example,

$$\text{arity}(\tau_{12}) = \text{Prime}_{01}, \quad \text{arity}(\tau_{10}) = \text{Prime}_{10}.$$

This single example already exhibits two different difficulties which arise when we try to analyze the comparison between two distinct 2-preptyx terms.

First consider $\tau_\emptyset$ and τ_0 . The priority predendrogram of $\tau_\emptyset$ consists of a terminal root, while the corresponding root in the priority predendrogram of τ_0 is non-terminal. Hence there is no ordinary diagram in which these two roots are identified: an ordinary predendrogram morphism must preserve terminal nodes.

Nevertheless, this is precisely an identification which we want to consider when analyzing the comparison of the two terms. If $D \models x_0 \multimap x_1$ and x_0 is a root, then x_0 cannot literally be substituted into the nullary term $\tau_\emptyset$, since it is non-terminal. By temporarily

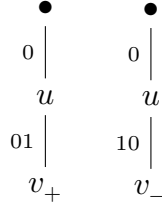


Figure 4.6: Arities of τ_{12} and τ_{10}

ignoring terminality, however, we may regard the two terms as if their first priority nodes agreed. Passing to the completion of D then supplies a hidden terminal copy of x_0 , and the resulting comparison is well-defined. This is the basic motivation for introducing *ostensible diagrams*.

There is, however, a second phenomenon which ostensible diagrams do not remove. Consider the two terms τ_{12} and τ_{10} . Both first inspect the argument 1. Figure 4.6 illustrates the arity of τ_{12} and τ_{10} respectively, which are Prime_{01} and Prime_{10} respectively. The roots have empty priority order, and the two copies of the node of level 1 have the unique priority order on one element. At the next level, however,

$$\pi^{\text{Prime}_{01}}(v_+) = (0 < 1), \quad \pi^{\text{Prime}_{10}}(v_-) = (1 < 0).$$

We can identify the roots and realize the two copies of u as distinct immediate successors of the common root. Their local order may be chosen in either direction. Consequently, after the first priority node has been identified, the comparison between τ_{12} and τ_{10} can still vary in both directions.

On the other hand, the two copies of u cannot themselves be identified, even in an ostensible diagram. If they were identified, the images of v_+ and v_- would be immediate successors of the same node, and hence would have to carry the same priority order. This contradicts $(0 < 1) \neq (1 < 0)$.

Thus two different obstructions have to be distinguished. In the pair $\tau_\emptyset, \tau_0$, the desired agreement fails only because one of the corresponding nodes is terminal, and it can be

recovered by passing to an ostensible diagram and then to a completion. In the pair τ_{12}, τ_{10} , the agreement cannot be extended because the two terms demand incompatible continuation priority orders.

The previous example shows that it is not enough to ask for the least p such that agreement of the first p arguments gives a fixed comparison. We must also keep track of whether such an agreement can be realized at all. Accordingly, for two terms s, t and a natural number p , we will consider the set of comparison directions realized by p -congruent ostensible diagrams. Such a set may have two elements, one element, or no elements:

- two elements mean that the comparison can still be changed in either direction;
- one element means that the comparison has stabilized;
- the empty set means that the prescribed ostensible agreement is itself impossible.

The comparison depth $\mathbf{p}^P(s, t)$ will be defined as the first stage at which one of the latter two situations occurs.

Now we define ostensible diagrams as we promised.

Definition 4.6.2. An *ostensible 1-arity diagram* is a diagram of the form

$$\mathbb{D} = \begin{array}{ccc} d_{\cap} & \dashrightarrow & d_1 \\ \vdots & & \vdots \\ d_0 & \dashrightarrow_{e_0} & d_{\cup} \end{array} \quad (4.12)$$

such that every predendrogram occurring $\mathbb{D}$ is natural, and $\mathbb{D}$ is a pullback and $\text{ran } e_0 \cup \text{ran } e_1 = d_{\cup}$, but both e_0 and e_1 can be quasimorphisms. (In particular, e_0 and e_1 may not respect terminal points. For two terms s, t of a 2-preptyx P , we say $\mathbb{D}$ is an *ostensible 1-arity diagram for s and t* if $d_0 = \text{arity } s$ and $d_1 = \text{arity } t$.

Similarly, we define an *ostensible 1-IU diagram* by mimicking [Definition 4.1.3](#), but we allow every morphism to be a quasimorphism, with the same pullback and union conditions, but allowing every predendrogram morphism to be a quasimorphism.

Definition 4.6.3. Let D be a predendrogram and $a, b \subseteq D$ be predecessor-closed subsets of D (That is, if $x \in a$ and $y \multimap x$, then $y \in a$, and the same for b .) We define $\text{Diag}_D(a, b)$ as an ostensible 1-arity diagram isomorphic to the following commutative diagram:

$$\mathbb{D} = \begin{array}{ccc} a \cap b & \xrightarrow{\subseteq} & b \\ \downarrow \subseteq & & \downarrow \subseteq \\ a & \xrightarrow{\subseteq} & a \cup b \end{array}$$

Similarly, for a predendrogram D and predecessor-closed subsets $a_0, \dots, a_{m-1} \subseteq D$, we define $\text{Diag}_D(a_0, \dots, a_{m-1})$ as an ostensible 1-IU diagram isomorphic to the inclusion diagram generated by $a_0, \dots, a_{m-1}$.

Then we unravel the hidden points in an ostensible 1-arity diagram by taking the completion:

Definition 4.6.4. For an ostensible 1-arity diagram (4.12), we define a new arity diagram $\mathbb{D}^*$ by

$$\mathbb{D}^* = \text{Diag}_c(\text{ran}(e_0^* \circ \text{cem}_{d_0}), \text{ran}(e_1^* \circ \text{cem}_{d_1}))$$

where $c = \text{ran}(e_0^* \circ \text{cem}_{d_0}) \cup \text{ran}(e_1^* \circ \text{cem}_{d_1})$. Similarly, for an ostensible 1-IU diagram $\mathbb{D}$ for m objects, we define

$$\mathbb{D}^* = \text{Diag}_{R_0 \cup \dots \cup R_{m-1}}(R_0, \dots, R_{m-1}),$$

where $\top = 0 \vee 1 \vee \dots \vee (m-1)$ and

$$R_i = \text{ran}(\mathbb{D}(i, \top)^* \circ \text{cem}_{\mathbb{D}(i)}) \subseteq \mathbb{D}(\top)^*.$$

Note that $\mathbb{D}^*$ is an ordinary 1-IU diagram. Indeed, for each $i < m$,

$$\mathbb{D}(i, \top)^* \circ \text{cem}_{\mathbb{D}(i)} : \mathbb{D}(i) \longrightarrow \mathbb{D}(\top)^*$$

is an ordinary predendrogram morphism. Hence, after identifying $\mathbb{D}(i)$ with its range R_i , the inclusion $R_i \rightarrow \bigcup_{j < m} R_j$ is again an ordinary predendrogram morphism. In particular, if a point is terminal in R_i , then it is already terminal in the ambient predendrogram

$\mathbb{A}(\top)^*$, so adjoining the other ranges cannot introduce a new successor below it. Thus the inclusion diagram generated by $R_0, \dots, R_{m-1}$ satisfies the requirements of a 1-IU diagram.

Lemma 4.6.5. *Let $\mathbb{A}$ be an ostensible 1-IU diagram with m objects. For any $i, j < m$, $\mathbb{A}^*[i, j] = (\mathbb{A}[i, j])^*$.*

Proof. Again, let $\top = 0 \vee \dots \vee (m-1)$. Write

$$S_i = \text{ran}(\mathbb{A}(i, i \vee j)^* \circ \text{cem}_{\mathbb{A}(i)}), \quad S_j = \text{ran}(\mathbb{A}(j, i \vee j)^* \circ \text{cem}_{\mathbb{A}(j)}).$$

Then $(\mathbb{A}[i, j])^* = \text{Diag}_{S_i \cup S_j}(S_i, S_j)$. On the other hand, if

$$R_\ell = \text{ran}(\mathbb{A}(\ell, \top)^* \circ \text{cem}_{\mathbb{A}(\ell)}) \quad (\ell < m),$$

then $\mathbb{A}^*[i, j] = \text{Diag}_{R_i \cup R_j}(R_i, R_j)$. By functoriality of completion,

$$\mathbb{A}(i \vee j, \top)^* \circ \mathbb{A}(i, i \vee j)^* \circ \text{cem}_{\mathbb{A}(i)} = \mathbb{A}(i, \top)^* \circ \text{cem}_{\mathbb{A}(i)},$$

and similarly for j . Hence the predendrogram embedding $\mathbb{A}(i \vee j, \top)^*$ maps S_i onto R_i and S_j onto R_j . It induces an isomorphism

$$\text{Diag}_{S_i \cup S_j}(S_i, S_j) \cong \text{Diag}_{R_i \cup R_j}(R_i, R_j),$$

and isomorphic 1-arity diagrams are the same, as desired. $\square$

Then we have the following.

Proposition 4.6.6. *Let P be a 2-preptyx, $s, t, u \in \text{field}(P)$, and $\mathbb{A}$ be an ostensible 1-IU diagram fwith the same pullback and union conditions, but allowing every predendrogram morphism to be a quasimorphism. or s, t , and u . If $P \models s <_{\mathbb{A}[0,1]^*} t <_{\mathbb{A}[1,2]^*} u$, then $P \models s <_{\mathbb{A}[0,2]^*} u$.*

Proof. By Lemma 4.6.5, $\mathbb{D}^*[0, 1] = \mathbb{D}[0, 1]^*$ and $\mathbb{D}^*[1, 2] = \mathbb{D}[1, 2]^*$. Hence $P \models s <_{\mathbb{D}^*[0, 1]} t <_{\mathbb{D}^*[1, 2]} u$. Since $\mathbb{D}^*$ is an ordinary 1-IU diagram, the transitivity condition for P gives $P \models s <_{\mathbb{D}^*[0, 2]} u$. Again by Lemma 4.6.5, $\mathbb{D}^*[0, 2] = \mathbb{D}[0, 2]^*$, and so $P \models s <_{\mathbb{D}[0, 2]^*} u$. $\square$

Definition 4.6.7. Let P be a 2-preptyx and $s, t \in \text{field}(P)$. Let $\hat{p}_{s,t} \leq \min(|\pi^P(s)|, |\pi^P(t)|)$ be the largest natural number such that

1. For every $i, j < \hat{p}_{s,t}$, $\pi^P(s) \models i \multimap j$ iff $\pi^P(t) \models i \multimap j$. In particular, for every $i < \hat{p}_{s,t}$, $\text{lh}^{\pi^P(s)}(i) = \text{lh}^{\pi^P(t)}(i)$.
2. For every $i, j < \hat{p}_{s,t}$, $\pi^P(s) \models i < j$ iff $\pi^P(t) \models i < j$.
3. For every $i < \hat{p}_{s,t}$, $\pi^{\pi^P(s)}(i) = \pi^{\pi^P(t)}(i)$.

That is, $\hat{p}_{s,t}$ is the largest natural number p such that the restriction of $\pi^P(s)$ to p is equal to the restriction of $\pi^P(t)$ to p , possibly except the set of terminal points. We say an ostensible 1-arity diagram $\mathbb{D}$ for s and t is *p-congruent (relative to s and t)* if $e_0 \circ \text{en}_{\pi^P(s)}^{-1}(i) = e_1 \circ \text{en}_{\pi^P(t)}^{-1}(i)$ holds for every $i < p$. (Note that $d_0 \cong \pi^P(s)$ and $d_1 \cong \pi^P(t)$ by the assumption on $\mathbb{D}$.) Then we say $p \leq \hat{p}_{s,t}$ is *secure (relative to s and t)* if $P \models s <_{\mathbb{D}^*} t$ holds for some p -congruent ostensible $\mathbb{D}$, then $P \models s <_{\mathbb{D}^*} t$ holds for every p -congruent ostensible $\mathbb{D}$. In other words, the validity of $P \models s <_{\mathbb{D}^*} t$ does not depend on the choice of a p -congruent ostensible $\mathbb{D}$.

As we did in Section 2.3, we prove that a secure number relative to s and t exists by showing the following:

Theorem 4.6.8. $\hat{p}_{s,t}$ is secure.

Proof. For notational convenience, let us write $\pi_s = \pi_s^P$ and $\pi_t = \pi_t^P$. If $\hat{p}_{s,t} = |\pi_s| \leq |\pi_t|$, then we have a predendrogram quasimorphism $h: d_0 \rightarrow d_1$ given by

$$h(\text{en}_{\pi_s}^{-1}(i)) = \text{en}_{\pi_t}^{-1}(i).$$

Furthermore, if $\mathbb{D}$ is a $\hat{p}_{s,t}$ -congruent ostensible 1-arity diagram for s and t of the form (4.12), then we have $e_0 = e_1 \circ h$ and $e_0^* = e_1^* \circ h^*$. Then we have

$$\begin{aligned} P \models s <_{\mathbb{D}^*} t &\iff P(d_{\cup}^*) \models s(\text{ran } e_0^* \circ \text{cem}_{d_0}) < t(\text{ran } e_1^* \circ \text{cem}_{d_1}) \\ &\iff P(d_1^*) \models s(\text{ran } h^* \circ \text{cem}_{d_0}) < t(\text{ran } \text{cem}_{d_1}), \end{aligned}$$

and the last sentence is independent of a $\hat{p}_{s,t}$ -congruent $\mathbb{D}$. Hence $\hat{p}_{s,t}$ is secure relative to s and t . The case $\hat{p}_{s,t} = |\pi_t| \leq |\pi_s|$ is analogous, so let us consider the case $\hat{p}_{s,t} < \min(|\pi_s|, |\pi_t|)$. Note that $s \neq t$ in this case since $\hat{p}_{t,t} = |\pi_t|$. [The case where there is no \$\hat{p}_{s,t}\$ -congruent ostensible diagram is trivial, so we assume that there is some such diagram.](#) For notational convenience, let us write $\hat{p} = \hat{p}_{s,t}$. Now suppose that

$$\mathbb{D}' = \begin{array}{ccc} d'_{\cap} & \dashrightarrow & d_1 \\ \downarrow & & \downarrow e'_1 \\ d_0 & \dashrightarrow_{e'_0} & d'_{\cup} \end{array} \quad (4.13)$$

is another ostensible $\hat{p}$ -congruent 1-arity diagram with respect to s and t . We will claim that $P \models s <_{\mathbb{D}^*} t$ implies $P \models s <_{(\mathbb{D}')^*} t$. Observe that the $\hat{p}$ -congruence of $\mathbb{D}$ implies

$$\forall i < \hat{p} [e_0 \circ \text{en}_{\pi_s, d_0}(i) = e_1 \circ \text{en}_{\pi_t, d_1}(i)].$$

The next [claim](#) says we can join $d_{\cup}$ and $d'_{\cup}$ in a way that their intersection is isomorphic to a predendrogram isomorphic to $\pi_s \upharpoonright \hat{p} = \pi_t \upharpoonright \hat{p}$, and the $\hat{p}$ th nodes differ in size.

Claim. Let $C = \pi_s \upharpoonright \hat{p} = \pi_t \upharpoonright \hat{p}$. Then we can find a new predendrogram D and embeddings $h: d_{\cup} \rightarrow D$, $h': d'_{\cup} \rightarrow D$ such that the following diagram commutes. (Here, the dashed arrows are quasimorphisms, and not morphisms in general.)

$$\begin{array}{ccc} C & \dashrightarrow_{e'_0 \circ \text{en}_{\pi_s, d_0}} & d'_{\cup} \\ \downarrow_{e_0 \circ \text{en}_{\pi_s, d_0}} & & \downarrow_{h'} \\ d_{\cup} & \xrightarrow{h} & D \end{array}$$

Furthermore, h and h' satisfy

1. $D \models h'(e'_0 \circ \text{en}_{\pi_s, d_0}(\hat{p})) \leq h(e_0 \circ \text{en}_{\pi_s, d_0}(\hat{p}))$, and
2. $D \models h(e_1 \circ \text{en}_{\pi_t, d_1}(\hat{p})) \leq h'(e'_1 \circ \text{en}_{\pi_t, d_1}(\hat{p}))$.

Proof. Let $c = e_0 \circ \text{en}_{\pi_s, d_0} \upharpoonright C$ and $c' = e'_0 \circ \text{en}_{\pi_s, d_0} \upharpoonright C$. By congruence, these are also the maps induced by e_1 and e'_1 , respectively. We first verify that $c: C \rightarrow d_\cup$ and $c': C \rightarrow d'_\cup$ satisfy the compatibility condition for predendrogram amalgamation.

Let $i < \hat{p}$, and suppose $d_\cup \models c(i) \multimap x$ and $d'_\cup \models c'(i) \multimap x'$. Since $d_\cup = \text{ran}(e_0) \cup \text{ran}(e_1)$ and $d'_\cup = \text{ran}(e'_0) \cup \text{ran}(e'_1)$, the points x and x' arise from immediate successors of the i th nodes of π_s or π_t . If both π_s and π_t have immediate successors at the i th node, their priority orders must agree since $\mathbb{D}$ is $\hat{p}$ -congruent and their images are immediate successors of the same point of $d_\cup$. Hence, in every case, $\pi^{d_\cup}(x) = \pi^{d'_\cup}(x')$. Moreover, $c(i)$ is terminal in $d_\cup$ exactly when the i th nodes of both π_s and π_t are terminal. The same characterization holds for $c'(i)$ in $d'_\cup$. Thus c and c' also satisfy the terminal-point condition required for a proper amalgamation. Hence [Theorem 3.3.11](#) applies and gives an amalgamation in which the two copies of C are identified and the maps from $d_\cup$ and $d'_\cup$ are embeddings.

It remains to choose the local orders in the amalgamation. Set

- $a = e_0 \circ \text{en}_{\pi_s, d_0}(\hat{p})$, $b = e_1 \circ \text{en}_{\pi_t, d_1}(\hat{p})$, and
- $a' = e'_0 \circ \text{en}_{\pi_s, d_0}(\hat{p})$, $b' = e'_1 \circ \text{en}_{\pi_t, d_1}(\hat{p})$.

We describe more precisely the positions of these four points relative to the common copy of C . Suppose first that the $\hat{p}$ th node of π_s is a root. Then a and a' belong to the local linear orders of roots, and the roots belonging to C determine a initial segment in each of these orders. Since the embeddings of C agree, a and a' determine the same initial segment over C .

If the $\hat{p}$ th node of π_s is not a root, let $i < \hat{p}$ be its immediate predecessor. Then a and a' are immediate successors of the common image of the i th node, and the immediate

successors of this node which belong to the common copy of C form a common finite suborder. Again a and a' determine the same initial segment in this common local order. The analogous statement holds for b and b' .

We claim that, whenever a and b belong to the same local order, the initial segments determined by a and b over C are distinct. Indeed, suppose otherwise. Then the $\hat{p}$ th nodes of π_s and π_t are either both roots or have the same immediate predecessor, and they have the same comparison relation with every earlier node in the corresponding local order. Moreover, their priority orders agree: if they have the same immediate predecessor, this follows from the definition of a predendrogram, while if they are roots, both priority orders are the unique linear order on 0 . Hence π_s and π_t agree, up to terminality, on their first $\hat{p} + 1$ nodes, contrary to the maximality of $\hat{p}$.

Consequently, if the two pairs occur in the same local order, the initial segments determined by a, a' and by b, b' are distinct. If they occur in different local orders, the two choices below are independent. By inspection of the proof of [Theorem 3.3.11](#), the two amalgamands may be interleaved independently inside each initial segment of a common local order, subject only to preserving the local orders already present in the two amalgamands. We may therefore choose the amalgamation so that

$$h'(a') \leq h(a) \quad \text{and} \quad h(b) \leq h'(b'). \quad \square$$

We are ready to prove that $\hat{p}$ is secure. Given the amalgamation in the previous claim, we have

- $(h' \circ e'_0)(\text{en}_{\pi_s, d_0}(i)) = (h \circ e_0)(\text{en}_{\pi_s, d_0}(i))$ for every $i < \hat{p}$, and
- $(h' \circ e'_0)(\text{en}_{\pi_s, d_0}(\hat{p})) \leq (h \circ e_0)(\text{en}_{\pi_s, d_0}(\hat{p}))$.

This implies

- $(h' \circ e'_0)^* \circ \text{cem}_{d_0}(\text{en}_{\pi_s, d_0}(i)) = (h \circ e_0)^* \circ \text{cem}_{d_0}(\text{en}_{\pi_s, d_0}(i))$ for every $i < \hat{p}$, and
- $(h' \circ e'_0)^* \circ \text{cem}_{d_0}(\text{en}_{\pi_s, d_0}(\hat{p})) < (h \circ e_0)^* \circ \text{cem}_{d_0}(\text{en}_{\pi_s, d_0}(\hat{p}))$.

Hence by **Theorem 4.5.10**, we have

$$P(D^*) \models s(\text{ran}(h' \circ e'_0)^* \circ \text{cem}_{d_0}) < s(\text{ran}(h \circ e_0)^* \circ \text{cem}_{d_0}).$$

Similarly, we have

$$P(D^*) \models t(\text{ran}(h \circ e_1)^* \circ \text{cem}_{d_1}) < t(\text{ran}(h' \circ e'_1)^* \circ \text{cem}_{d_1})$$

Hence we have

$$\begin{aligned}
& P \models s <_{\mathbb{D}^*} t \\
& \iff P(d_{\cup}^*) \models s(\text{ran } e_0^* \circ \text{cem}_{d_0}) < t(\text{ran } e_1^* \circ \text{cem}_{d_1}) \\
& \iff P(D^*) \models s(\text{ran}(h \circ e_0)^* \circ \text{cem}_{d_0}) < t(\text{ran}(h \circ e_1)^* \circ \text{cem}_{d_1}) \\
& \implies P(D^*) \models s(\text{ran}(h' \circ e'_0)^* \circ \text{cem}_{d_0}) < t(\text{ran}(h' \circ e'_1)^* \circ \text{cem}_{d_1}) \\
& \iff P((d'_{\cup})^*) \models s(\text{ran}(e'_0)^* \circ \text{cem}_{d_0}) < t(\text{ran}(e'_1)^* \circ \text{cem}_{d_1}) \\
& \iff P \models s <_{(\mathbb{D}')^*} t.
\end{aligned}$$

Since $\mathbb{D}$ and $\mathbb{D}'$ are arbitrary $\hat{p}$ -congruent ostensible 1-arity diagrams, we have the desired result. $\square$

Hence, we can define the following:

Definition 4.6.9. For a 2-preptyx P and two different P -terms s and t , let $\mathbf{p}^P(s, t)$ be the least secure number relative to s and t . Then we define

- $\varepsilon_{s,t}^P := 0$ if there is no $\mathbf{p}^P(s, t)$ -congruent ostensible 1-arity diagram for s and t ,
- $\varepsilon_{s,t}^P := +1$ if $P \models s <_{\mathbb{D}^*} t$ holds for every (equivalently, some) $\mathbf{p}^P(s, t)$ -congruent ostensible 1-arity diagram $\mathbb{D}$ for s and t , and

- $\varepsilon_{s,t}^P := -1$ if $P \models s >_{\mathbb{A}^*} t$ holds for every (equivalently, some) $\mathbf{p}^P(s, t)$ -congruent ostensible 1-arity diagram $\mathbb{A}$ for s and t .

Note that, by definition of $\mathbf{p}^P(s, t)$, for $q < \mathbf{p}^P(s, t)$, we can find ostensible 1-arity diagrams $\mathbb{A}$ and $\mathbb{B}$ such that $P \models s <_{\mathbb{A}^*} t$ and $P \models t <_{\mathbb{B}^*} s$. Also, we can easily get

$$\mathbf{p}^P(s, t) = \mathbf{p}^P(t, s) \quad \text{and} \quad \varepsilon_{s,t}^P = -\varepsilon_{t,s}^P.$$

We also have $\mathbf{p}^P(s, s) = |\pi_s^P|$ by [Theorem 4.5.10](#). As like predilators, Σ^P , $\mathbf{p}^P$, and ε^P completely determine the structure of a 2-preptyx P .

Theorem 4.6.10. *Let P be a 2-preptyx and D be a predendrogram. For $s, t \in \text{field}(P)$ and $a, b \subseteq D$ with $a \cong \text{arity } s$, $b \cong \text{arity } t$, we have $P(D) \models s(a) < t(b)$ if and only if either*

1. $\text{en}_{\pi^{P(s),a}}(i) = \text{en}_{\pi^{P(t),b}}(i)$ for every $i < \mathbf{p}^P(s, t)$ and $\varepsilon_{s,t}^P = +1$, or
2. There is $j < \mathbf{p}^P(s, t)$ such that $\text{en}_{\pi^{P(s),a}}(i) = \text{en}_{\pi^{P(t),b}}(i)$ for every $i < j$ and $\text{en}_{\pi^{P(s),a}}(j) < \text{en}_{\pi^{P(t),b}}(j)$.

Proof. It suffices to prove the backward implication of the theorem. Observe that if the first $\mathbf{p}^P(s, t)$ arguments of a and b agree, then the diagram induced by a and b witnesses $\varepsilon_{s,t}^P = \pm 1$, so there is no need to consider the case $\varepsilon_{s,t}^P = 0$.

Suppose that the first disjunct holds. In this case, $\text{Diag}_D(a, b)$ is $\mathbf{p}^P(s, t)$ -congruent, so $\varepsilon_{s,t}^P = +1$ implies $P \models s <_{\text{Diag}_D(a,b)^*} t$, which implies $P(D^*) \models s(\text{ran cem}_a) < t(\text{ran cem}_b)$. We can see that ran cem_a and ran cem_b are images of a and b under $\text{cem}_D: D \rightarrow D^*$, so we get $P(D) \models s(a) < t(b)$.

Now suppose that the second disjunct holds. Then the witnessing j is not secure, so there is a j -congruent ostensible 1-arity diagram

$$\mathbb{A} = \begin{array}{ccc} d_{\cap} & \dashrightarrow & d_1 \\ \vdots & & \vdots e_1 \\ d_0 & \dashrightarrow_{e_0} & d_{\cup} \end{array} \quad (4.14)$$

for s and t such that $P \models s <_{\mathbb{A}^*} t$. Note that $\|a\| = \|\pi^P(s)\| = d_0$ and $\|b\| = \|\pi^P(t)\| = d_1$.

Claim. Let $C = \pi^P(s) \upharpoonright j$. We can find a finite predendrogram E and embeddings $h: a \cup b \rightarrow E$, $h': d_{\cup} \rightarrow E$ such that the following diagram commutes, where the dashed arrows are quasimorphisms:

$$\begin{array}{ccc}
 C & \xrightarrow{\text{en}_{\pi^P(s),a} \upharpoonright j} & a \cup b \\
 \downarrow e_0 \circ \text{en}_{\pi^P(s),d_0} \upharpoonright j & & \downarrow h' \\
 d_{\cup} & \xrightarrow{h} & E
 \end{array} \tag{4.15}$$

Furthermore, we have

- $E \models h'(\text{en}_{\pi^P(s),a}(j)) \leq h(e_0 \circ \text{en}_{\pi^P(s),d_0}(j))$,
- $E \models h(e_1 \circ \text{en}_{\pi^P(t),d_1}(j)) \leq h'(\text{en}_{\pi^P(t),b}(j))$.

Proof. Its proof is more or less similar to that of the Claim in the proof of [Theorem 4.6.8](#). We first claim that the two dashed arrows in the diagram (4.15) satisfy the hypotheses of [Theorem 3.3.11](#). Indeed, at every node of C , immediate successors coming from π_s and π_t have the same priority order whenever both exist, since both diagrams are j -congruent. Furthermore, the image of a node of C is terminal in either union exactly when the corresponding nodes of both π_s and π_t are terminal.

Now let

- $x_s = \text{en}_{\pi_s,a}(j)$, $x_t = \text{en}_{\pi_t,b}(j)$, and
- $y_s = e_0 \circ \text{en}_{\pi_s,d_0}(j)$, $y_t = e_1 \circ \text{en}_{\pi_t,d_1}(j)$.

We have $a \cup b \models x_s \leq x_t$ by the assumption. Also, since $j < \mathbf{p}^P(s, t)$, the four nodes x_s, x_t, y_s, y_t have the same initial segments of the corresponding local order over the common copy of C . By inspecting the proof of [Theorem 3.3.11](#), we can amalgamate $a \cup b$ and $d_{\cup}$ into E in a way that, if $h: a \cup b \rightarrow E$ and $h': d_{\cup} \rightarrow E$ amalgamate the two predendrogams, then $E \models h'(y_t) \leq h'(y_s)$ and $E \models h(x_s) \leq h(x_t)$. $\square$

Let us go back to the main proof. By the choice of j , h , and h' , we have the following two:

1. For every $i < j$, we have $h(e_0 \circ \text{en}_{\pi^P(s), d_0}(i)) = h'(\text{en}_{\pi^P(s), a}(i))$.
2. $h(e_0 \circ \text{en}_{\pi^P(s), d_0}(j)) > h'(\text{en}_{\pi^P(s), a}(j))$.

Then we can see that

$$(h \circ e_0)^*(\text{cem}_{d_0} \circ \text{en}_{\pi^P(s), d_0}(i)) = (h' \circ \text{en}_{d_0, a})^*(\text{cem}_{d_0} \circ \text{en}_{\pi^P(s), d_0}(i))$$

for $i < j$ as follows: For notational convenience, let $x = \text{en}_{\pi^P(s), d_0}(i)$. Then

$$\begin{aligned} & (h \circ e_0)^*(\text{cem}_{d_0}(x)) \\ &= (h_0 \circ e_0)^*(x) = \begin{cases} (h \circ e_0)(x), & x \in \text{term } d_0 \leftrightarrow (h \circ e_0)(x) \in \text{term } E \\ (h \circ e_0)(x)^*, & \text{otherwise.} \end{cases} \end{aligned}$$

Similarly,

$$\begin{aligned} & (h' \circ \text{en}_{d_0, a})^*(\text{cem}_{d_0}(x)) = (h' \circ \text{en}_{d_0, a})^*(x) \\ &= \begin{cases} (h' \circ \text{en}_{d_0, a})(x), & x \in \text{term } d_0 \leftrightarrow (h' \circ \text{en}_{d_0, a})(x) \in \text{term } E \\ (h' \circ \text{en}_{d_0, a})(x)^*, & \text{otherwise.} \end{cases} \end{aligned}$$

But we know that $(h \circ e_0)(x) = (h' \circ \text{en}_{d_0, a})(x)$. Hence, the conditionals in each equality coincide, and we get the desired equality. We also have

$$E^* \models (h \circ e_0)^*(\text{cem}_{d_0} \circ \text{en}_{\pi^P(s), d_0}(j)) > (h' \circ \text{en}_{d_0, a})^*(\text{cem}_{d_0} \circ \text{en}_{\pi^P(s), a}(j)),$$

so [Theorem 4.5.10](#) implies

$$P(E^*) \models s(\text{ran}((h')^* \circ \text{en}_{d_0, a}^* \circ \text{cem}_{d_0})) < s(\text{ran}(h^* \circ e_0^* \circ \text{cem}_{d_0})).$$

By a similar argument, we also have

$$P(E^*) \models t(\text{ran}(h^* \circ e_1^* \circ \text{cem}_{d_1})) < t(\text{ran}((h')^* \circ \text{en}_{d_1, b}^* \circ \text{cem}_{d_1})).$$

From $P \models s <_{\mathbb{D}^*} t$, we have

$$P(E^*) \models s(h^*[\text{ran } e_0^* \circ \text{cem}_{d_0}]) < t(h^*[\text{ran } e_1^* \circ \text{cem}_{d_1}]).$$

In sum, we can deduce

$$P(E^*) \models s(\text{ran}((h')^* \circ \text{en}_{d_0,a}^* \circ \text{cem}_{d_0})) < t(\text{ran}((h')^* \circ \text{en}_{d_1,b}^* \circ \text{cem}_{d_1})).$$

This readily implies

$$P((a \cup b)^*) \models s(\text{ran}(\text{en}_{d_0,a}^* \circ \text{cem}_{d_0})) < t(\text{ran}(\text{en}_{d_1,b}^* \circ \text{cem}_{d_1})).$$

We have $\text{ran}(\text{en}_{d_0,a}^* \circ \text{cem}_{d_0}) = \text{ran } \text{cem}_a = \text{cem}_{a \cup b}[a]$ and $\text{ran}(\text{en}_{d_1,b}^* \circ \text{cem}_{d_1}) = \text{ran } \text{cem}_b = \text{cem}_{a \cup b}[b]$. Hence we have $P(a \cup b) \models s(a) < t(b)$, which is equivalent to $P(D) \models s(a) < t(b)$. $\square$

The following lemma has a focal role in the latter proofs.

Lemma 4.6.11. *Let P be a 2-preptyx, let $s, t \in \text{field}(P)$, and let $q < \mathbf{p}^P(s, t)$. If $\mathbb{D}$ is a q -congruent ostensible 1-arity diagram for s and t , then $\mathbb{D}^*$ is q -congruent. More explicitly, if*

$$\mathbb{D} = \begin{array}{ccc} d_{\cap} & \dashrightarrow & d_1 \\ \downarrow & & \downarrow e_1 \\ d_0 & \dashrightarrow_{e_0} & d_{\cup} \end{array},$$

then, for every $i < q$,

$$(e_0^* \circ \text{cem}_{d_0})(\text{en}_{\pi_s^P, d_0}(i)) = (e_1^* \circ \text{cem}_{d_1})(\text{en}_{\pi_t^P, d_1}(i)). \quad (4.16)$$

Proof. Suppose otherwise, and let $i < q$ be least such that

$$(e_0^* \circ \text{cem}_{d_0})(\text{en}_{\pi_s^P, d_0}(i)) \neq (e_1^* \circ \text{cem}_{d_1})(\text{en}_{\pi_t^P, d_1}(i)).$$

Since $\mathbb{D}$ is q -congruent, before taking the completion we have

$$y := e_0(\text{en}_{\pi_s^P, d_0}(i)) = e_1(\text{en}_{\pi_t^P, d_1}(i)).$$

Thus, the two completed images can be different only because the corresponding i -th nodes of π_s^P and π_t^P have different terminal status. More precisely, one of these nodes is terminal in its arity, and the other is nonterminal. In the completion, the nonterminal occurrence is sent to the original point y , whereas the terminal occurrence is sent to its terminal copy y^* . By the definition of completion, $y < y^*$. Consequently, the order between the two completed i th arguments is determined solely by which of the two corresponding nodes is terminal. Notice that this terminality information belongs to the arities π_s^P and π_t^P themselves and is therefore independent of the choice of an ostensible arity diagram. Moreover, by the minimality of i , for every $j < i$ we have

$$(e_0^* \circ \text{cem}_{d_0})(\text{en}_{\pi_s^P, d_0}(j)) = (e_1^* \circ \text{cem}_{d_1})(\text{en}_{\pi_t^P, d_1}(j)).$$

Since $\mathbb{D}$ is q -congruent, the corresponding nodes before completion already have the same image. Hence the corresponding j th nodes of π_s^P and π_t^P must have the same terminal status for every $j < i$.

Now let $\mathbb{K}$ be any $(i+1)$ -congruent ostensible 1-arity diagram for s and t . For every $j < i$, the corresponding j th nodes have the same terminal status, so their common image in $\mathbb{K}$ remains common after completion. At the index i , however, the corresponding nodes have different terminal status. Hence their common image, say z , is separated by completion into z and z^* , with $z < z^*$. Therefore, the first difference between the completed arguments of s and t occurs precisely at index i , and its direction is the same for every $(i+1)$ -congruent ostensible 1-arity diagram.

It follows from [Theorem 4.6.10](#) that the comparison between s and t is independent of the choice of an $(i+1)$ -congruent ostensible 1-arity diagram. Hence $i+1$ is secure relative to s and t . By the definition of $\mathbf{p}^P(s, t)$ as the least secure number, $\mathbf{p}^P(s, t) \leq i+1$. On the other hand, $i+1 \leq q < \mathbf{p}^P(s, t)$, which is a contradiction. $\square$

The following proposition characterizes the case $\varepsilon_{s,t}^P = 0$.

Proposition 4.6.12. *Let P be a 2-preptyx, $s, t \in P$. Then we have the following:*

1. *If $\varepsilon_{s,t}^P = 0$, then $\mathbf{p}^P(s, t) > 0$.*
2. *Now suppose that $p = \mathbf{p}^P(s, t) > 0$. Then the following are equivalent:*
 - (a) $\varepsilon_{s,t}^P = 0$.
 - (b) *In both of π_s^P and π_t^P , $(p-1)$ is a non-terminal node and $\text{spr}^{\pi_s^P}(p-1) \neq \text{spr}^{\pi_t^P}(p-1)$.*

Here, the successor priority $\text{spr}^D(x)$ for a predendrogram D and a non-terminal node $x \in D$ is π_y^D for any extension $y \circ x$.

Proof. 1. Suppose that $\mathbf{p}^P(s, t) = 0$. We can find an ostensible 1-arity diagram with empty pullback, so $\varepsilon_{s,t}^P \neq 0$.

2. For the right-to-left direction, we cannot amalgamate π_s^P and π_t^P with two copies of $(p-1)$ agree; otherwise the common image of $(p-1)$ would have immediate successors coming from the two arities with different priority orders, contradicting the predendrogram axioms. For the left-to-right direction, we prove the contrapositive. We have either

- $(p-1)$ is a terminal in at least one of π_s^P or π_t^P , or
- $(p-1)$ is not a terminal, and $\text{spr}^{\pi_s^P}(p-1) = \text{spr}^{\pi_t^P}(p-1)$.

We have the amalgamation conditions for $i < p-1$ both in π_s^P and π_t^P : If $i < p-1$, then $i+1 < p = \mathbf{p}^P(s, t)$ is not secure, so we can find an $(i+1)$ -congruent ostensible 1-arity diagram. This diagram identifies two copies of i from π_s^P and π_t^P , so they have the same successor priority dendrograms if i is non-terminal in both π_s^P and π_t^P . Also, in either case, we also have the amalgamation condition for $(p-1)$. Combined them with $\mathbf{p}^P(s, t) \leq \hat{p}_{s,t}$, we can amalgamate π_s^P and π_t^P up to p to form an ostensible 1-arity diagram witnessing $\varepsilon_{s,t}^P = \pm 1$. $\square$

Like the predilator case, $\mathbf{p}^P$ has similar constraints.

Proposition 4.6.13. *Let P be a 2-preptyx. For $s, t, u \in \text{field}(P)$, we have $\mathbf{p}^P(s, u) \geq \min(\mathbf{p}^P(s, t), \mathbf{p}^P(t, u))$.*

Proof. Put $r = \min(\mathbf{p}^P(s, t), \mathbf{p}^P(t, u))$, and let $p < r$. We show that p is not secure relative to s and u . Since $p < \mathbf{p}^P(s, t)$, p is not secure relative to s and t . Hence there are p -congruent ostensible 1-arity diagrams $\mathbb{D}_+$ and $\mathbb{D}_-$ such that $P \models s <_{\mathbb{D}_+} t$ and $P \models t <_{\mathbb{D}_-} s$. Similarly, since $p < \mathbf{p}^P(t, u)$, there are p -congruent ostensible 1-arity diagrams $\mathbb{K}_+$ and $\mathbb{K}_-$ such that $P \models t <_{\mathbb{K}_+} u$ and $P \models u <_{\mathbb{K}_-} t$. By Lemma 4.6.11, the completed diagrams $\mathbb{D}_+^*$ and $\mathbb{K}_+^*$ are still p -congruent. Moreover, they are ordinary 1-arity diagrams. Hence we can consider the following amalgamation maps f_0 and f_1 making the following diagram commute:

$$\begin{array}{ccc} (\text{arity } t)^* & \xrightarrow{\mathbb{D}_+^*(1, 0 \vee 1)} & \mathbb{K}_+^*(0 \vee 1) \\ \mathbb{K}_+^*(0, 0 \vee 1) \downarrow & & \downarrow f_1 \\ \mathbb{D}_+^*(0 \vee 1) & \xrightarrow{f_0} & D \end{array}$$

Note that $\mathbb{D}(1) = \mathbb{K}(0) = \text{arity } t$. Then consider the ordinary 1-IU diagram

$$\mathbb{D}_+ = \text{Diag}_D(f_0[\mathbb{D}_+^*(0)], f_0[\mathbb{D}_+^*(1)], f_1[\mathbb{K}_+^*(1)]).$$

By construction, $\mathbb{D}^+[0, 1] = \mathbb{D}_+$ and $\mathbb{D}^+[1, 2] = \mathbb{K}_+$. Hence

$$P \models s <_{\mathbb{D}^+[0, 1]} t <_{\mathbb{D}^+[1, 2]} u,$$

so we get $P \models s <_{\mathbb{D}^+[0, 2]} u$.

Now we claim that $\mathbb{D}^+[0, 2]$ is p -congruent. Let $i < p$. Since $\mathbb{D}_+^*$ is p -congruent, the i th nodes of π_s^P and π_t^P have the same image in $\mathbb{D}_+(0 \vee 1)$. Likewise, since $\mathbb{K}_+^*$ is p -congruent, the i th nodes of π_t^P and π_u^P have the same image in $\mathbb{K}_+(0 \vee 1)$. The two copies of the i th node of π_t^P are identified in the amalgamation over $\text{arity}(t)$. Hence the i th nodes of π_s^P and π_u^P have the same image in D . This proves the claim.

Applying exactly the same construction to $\mathbb{A}_-^*$ and $\mathbb{X}_-^*$, we obtain a 1-IU diagram $\mathbb{A}_-$ with three objects such that $\mathbb{A}_-[0, 2]$ is a p -congruent ordinary 1-arity diagram for s and u , and $P \models u <_{\mathbb{A}_-[0,2]} s$. This shows that p is not secure relative to s and u . Thus $p < \mathbf{p}^P(s, u)$ for every $p < r$, as desired. $\square$

Now let us examine the properties of ε^P . Unlike the dilator case, the possibility $\varepsilon^P = 0$ makes the relation given by

$$s <^P t \iff \varepsilon_{s,t}^P = +1$$

no longer a partial order. In particular, the relation $<^P$ is no longer transitive. However, we will see later that we can still extend $<^P$ into a linear order by showing that this relation has no cycles.

Proposition 4.6.14. *For $s, t, u \in \text{field}(P)$, suppose that $\varepsilon_{s,t}^P = \varepsilon_{t,u}^P = +1$. Then $\varepsilon_{s,u}^P \neq -1$ and $\mathbf{p}^P(s, u) = \min(\mathbf{p}^P(s, t), \mathbf{p}^P(t, u))$.*

Proof. One direction of the inequality was shown in ???. Hence we prove from the assumption that $\varepsilon_{s,u}^D = +1$ and $\mathbf{p}^P(s, u) \leq \min(\mathbf{p}^P(s, t), \mathbf{p}^P(t, u))$ hold.

Let $p = \min(\mathbf{p}^P(s, t), \mathbf{p}^P(t, u))$. For a p -congruent ostensible 1-arity diagram $\mathbb{A}$ for s and u , we claim that $P \models s <_{\mathbb{A}^*} u$, proving both the inequality and $\varepsilon_{s,u}^P = +1$. To ease the proof, we move our attention from P to $P(\mathbb{Q}_1^*)$ as follows: Let $\mathbb{A}$ be a p -congruent ostensible 1-arity diagram of the form (4.14), and $\iota: d_{\cup} \rightarrow \mathbb{Q}_1$ be a predendrogram morphism. Then $P \models s <_{\mathbb{A}^*} u$ is equivalent to

$$P(\mathbb{Q}_1^*) \models s(\text{ran}(\iota \circ e_0)^* \circ \text{cem}_{d_0}) < u(\text{ran}(\iota \circ e_1)^* \circ \text{cem}_{d_1}).$$

Now let $a = \text{ran}(\iota \circ e_0)$ and $c = \text{ran}(\iota \circ e_1)$. a and c may not be subpredendrograms of $\mathbb{Q}_1$ since their terminal points may not be terminal points of $\mathbb{Q}_1$. Still, they form subsets closed under predecessors. If we let $\iota_a: a \rightarrow \mathbb{Q}_1$ and $\iota_c: c \rightarrow \mathbb{Q}_1$ be the inclusion quasimorphisms, then $P \models s <_{\mathbb{A}^*} u$ becomes equivalent to

$$P(\mathbb{Q}_1^*) \models s(\text{ran } \iota_a^* \circ \text{cem}_a) < u(\text{ran } \iota_c^* \circ \text{cem}_c).$$

Moreover, we also have $\text{en}_{\pi^P(s),a}(i) = \text{en}_{\pi^P(u),c}(i)$ for every $i < p \leq \mathbf{p}^P(s, u)$. Now, let us divide the cases:

1. Suppose that $p = \mathbf{p}^P(s, t) = \mathbf{p}^P(t, u)$. By the ω -homogeneity of $\mathbb{Q}_1$, we can find a predecessor-closed finite subset $b \subseteq \mathbb{Q}_1$ such that $b \cong \text{arity } t$ and $\text{en}_{\pi^P(s),a}(i) = \text{en}_{\pi^P(t),b}(i)$ for every $i < p$. Hence $\text{Diag}_{\mathbb{Q}_1}(a, b)$ and $\text{Diag}_{\mathbb{Q}_1}(b, c)$ become p -congruent ostensible 1-arity diagram. Hence **Theorem 4.6.10** with the assumption $\varepsilon_{s,t}^P = \varepsilon_{t,u}^P = +1$ implies

$$P(\mathbb{Q}_1^*) \models s(\text{ran } \iota_a^* \circ \text{cem}_a) < t(\text{ran } \iota_b^* \circ \text{cem}_b) < u(\text{ran } \iota_c^* \circ \text{cem}_c). \quad (4.17)$$

2. Now suppose that $p = \mathbf{p}^P(s, t) < \mathbf{p}^P(t, u)$. By the ω -homogeneity argument, let us find a predecessor-closed finite subset $b \subseteq \mathbb{Q}_1$ isomorphic to $\text{arity } t$ such that $\text{en}_{\pi^P(s),a}(i) = \text{en}_{\pi^P(t),b}(i)$ for every $i < p$ and $\text{en}_{\pi^P(t),b}(p) < \text{en}_{\pi^P(u),c}(p)$ over $\mathbb{Q}_1$. (4.17) still holds in this case; The first inequality is witnessed by that $\text{Diag}_{\mathbb{Q}_1}(a, b)$ is a $\mathbf{p}^P(s, t)$ -congruent ostensible 1-arity diagram and $\varepsilon_{s,t}^P = +1$. The second inequality is witnessed by the second case of **Theorem 4.6.10**.
3. Now suppose that $\mathbf{p}^P(s, t) > \mathbf{p}^P(t, u) = p$. We can find a predecessor-closed finite subset $b \subseteq \mathbb{Q}_1$ isomorphic to $\text{arity } t$ such that $\text{en}_{\pi^P(t),b}(i) = \text{en}_{\pi^P(u),c}(i)$ for every $i < p$ and $\text{en}_{\pi^P(s),a}(p) < \text{en}_{\pi^P(t),b}(p)$. We again have (4.17).

Hence, in all possible cases, we get the inequality. $\square$

The ultrametric $\mathbf{p}^P$ allows us to classify terms into cells. We use the following equivalence relation:

Definition 4.6.15. Let m be a natural number, P a 2-preptyx, and $s, t \in P$. We define

$$s \equiv_m^P t \iff [s = t \vee \mathbf{p}^P(s, t) > m].$$

One can easily see that $\equiv_m$ is an equivalence relation; the transitivity of $\equiv_m$ follows from the triangle inequality. These equivalence classes will later form the cells in the

2-predendrogram representation. The next proposition shows that the comparison data of such a class with terms outside it do not depend on the choice of representative.

Proposition 4.6.16. *Suppose that P is a 2-preptyx, $s, t, u \in P$, and $\mathbf{p}^P(s, t) > \mathbf{p}^P(s, u)$. Then $\mathbf{p}^P(t, u) = \mathbf{p}^P(s, u)$ and $\varepsilon_{t,u}^P = \varepsilon_{s,u}^P$.*

Proof. Let $q = \mathbf{p}^P(s, u)$. Then we have

$$\mathbf{p}^P(t, u) \geq \min(\mathbf{p}^P(t, s), \mathbf{p}^P(s, u)) = q.$$

If $\mathbf{p}^P(t, u) > q$, then

$$q = \mathbf{p}^P(s, u) \geq \min(\mathbf{p}^P(s, t), \mathbf{p}^P(t, u)) > q,$$

which is impossible, so $\mathbf{p}^P(t, u) = q$. Now we claim that

$$\varepsilon_{t,u}^P = 0 \iff \varepsilon_{s,u}^P = 0$$

This is clear when $q = 0$, so suppose that $q > 0$. From $\mathbf{p}^P(s, t) > q$, $(q - 1)$ have the same terminality in π_s^P and π_t^P . Moreover, if $(q - 1)$ is not terminal in π_s^P and π_t^P , then $\text{spr}^{\pi_s^P}(q - 1) = \text{spr}^{\pi_t^P}(q - 1)$. By [Proposition 4.6.12](#), we have that $\varepsilon_{s,u}^P = 0$ iff $(q - 1)$ is not terminal both in π_s^P and π_u^P , and $\text{spr}^{\pi_s^P}(q - 1) \neq \text{spr}^{\pi_u^P}(q - 1)$. Now we can see that this implies that $(q - 1)$ is not terminal π_t^P , and $\text{spr}^{\pi_t^P}(q - 1) \neq \text{spr}^{\pi_u^P}(q - 1)$. The other direction of the implication is analogous.

It remains to show that if $\varepsilon_{t,u}^P \neq 0$, then $\varepsilon_{t,u}^P = +1$ iff $\varepsilon_{s,u}^P = +1$. If $\varepsilon_{t,u}^P = +1$ but $\varepsilon_{s,u}^P = -1$, so we have $\varepsilon_{u,s}^P = +1$, then [Proposition 4.6.14](#) implies

$$q < \mathbf{p}^P(t, s) = \min(\mathbf{p}^P(t, u), \mathbf{p}^P(u, s)) = q,$$

a contradiction. The other implication follows similarly. $\square$

We can state [\(4.6.16\)](#) alternatively as follows: If $s \equiv_m^P t$ and $s \not\equiv_m^P u$, then $\mathbf{p}^P(s, u) = \mathbf{p}^P(t, u) \leq m$ and $\varepsilon_{s,u}^P = \varepsilon_{t,u}^P$. In particular, both $\mathbf{p}^P(s, u)$ and $\varepsilon_{s,u}^P$ are invariant for $\equiv_m^P$ -equivalent terms. Hence, for two $\equiv_m^P$ -equivalence classes $[s]_{\equiv_m^P}$ and $[t]_{\equiv_m^P}$, we can define

- $\mathbf{p}^P([s]_{\equiv_m^P}, [t]_{\equiv_m^P}) = \mathbf{p}^P(s, t)$, and
- $\varepsilon_{[s]_{\equiv_m^P}, [t]_{\equiv_m^P}}^P = \varepsilon_{s, t}^P$.

The properties described in this section are precisely what is needed to define *cell decomposition of 2-pretykes*. Unlike Girard's original treatment, we shall not introduce a separate abstract construction since the resulting cell decomposition provides a more controllable combinatorial representation directly. The properties we have secured are enough to describe the tree structure of 2-pretykes, but we need more examination to determine the structure of children of a parent.

The next proposition describes how much of the priority predendrogram is determined by $\equiv_m^P$. If two distinct terms are $\equiv_m^P$ -equivalent, then their priority predendrograms agree through node m after terminality is ignored; moreover, their terminalities agree strictly below node m .

Proposition 4.6.17. *Suppose that $s, t \in P$, $s \equiv_m^P t$, and $s \neq t$. Then, for every $i, j \leq m$,*

1. $\pi_s^P \models i \multimap j$ iff $\pi_t^P \models i \multimap j$,
2. $\pi_s^P \models i < j$ iff $\pi_t^P \models i < j$, and
3. $\pi_i^{\pi_s^P} = \pi_i^{\pi_t^P}$.

Moreover, for every $i < m$, the node i is terminal in π_s^P iff it is terminal in π_t^P .

Proof. Since $s \equiv_m^P t$ and $s \neq t$, we have $m < \mathbf{p}^P(s, t) \leq \hat{p}_{s, t}$. Hence (1)–(3) follow immediately from the definition of $\hat{p}_{s, t}$.

It remains to prove the assertion about terminality. Suppose, towards a contradiction, that the terminalities disagree below node m , and let $i < m$ be the least index at which they disagree. Since $i + 1 \leq m < \mathbf{p}^P(s, t)$, the number $(i + 1)$ is not secure, and hence there exists an $(i + 1)$ -congruent ostensible 1-arity diagram $\mathbb{D}$ for s and t .

In $\mathbb{D}$, let x be the common image of the i -th nodes, namely,

$$x = \mathbb{D}(0, 0 \vee 1) \circ \text{en}_{\pi_s^P, \text{arity } s}(i) = \mathbb{D}(1, 0 \vee 1) \circ \text{en}_{\pi_t^P, \text{arity } t}(i).$$

By the minimality of i , completion preserves the agreement of all preceding nodes. At node i , however, one occurrence is terminal, and the other is nonterminal. Hence completion represents the nonterminal occurrence by x and the terminal occurrence by x^* , with $x < x^*$. The resulting direction depends only on which of π_s^P and π_t^P has the terminal i -th node, and is therefore the same in every $(i+1)$ -congruent ostensible 1-arity diagram. Thus $(i+1)$ is secure relative to s and t , a contradiction. $\square$

We now examine the comparison structure between the children of a fixed equivalence class. Although the cell decomposition will be formally introduced later, it is convenient to call an $\equiv_m^P$ -equivalence class contained in an $\equiv_{m-1}^P$ -equivalence class its *child*. Thus, if A and B are distinct children of the same $\equiv_{m-1}^P$ -equivalence class and $s \in A, t \in B$, then $\mathbf{p}^P(s, t) = m$. Indeed, since s and t belong to the same $\equiv_{m-1}^P$ -class, we have $\mathbf{p}^P(s, t) > m-1$, while $A \neq B$ implies $\mathbf{p}^P(s, t) \leq m$.

The following observation allows us to classify the children according to the behavior of the $(m-1)$ -st node of their priority predendrograms.

Proposition 4.6.18. *The following information is invariant for $\equiv_m^P$ -equivalent terms $s \in P$, where $m > 0$:*

1. *Whether $(m-1)$ is terminal in π_s^P ;*
2. *If $(m-1)$ is nonterminal in π_s^P , the successor priority $\text{spr}^{\pi_s^P}(m-1)$.*

Proof. We can only consider the case when $s \neq t$ and $\mathbf{p}^P(s, t) > m$. The terminality of $(m-1)$ coincides between π_s^P and π_t^P by [Proposition 4.6.17](#). Now suppose that $(m-1)$ is nonterminal in π_s^P and π_t^P . In an m -congruent ostensible 1-arity diagram, which exists since $m < \mathbf{p}^P(s, t)$, the images of the $(m-1)$ -st nodes coincide. Since both nodes are nonterminal, their immediate successors are mapped to children of this common image. By the defining condition on priorities of siblings in a predendrogram, these children have the same priority. Therefore $\text{spr}^{\pi_s^P}(m-1) = \text{spr}^{\pi_t^P}(m-1)$. $\square$

Fix an $\equiv_{m-1}^P$ -equivalence class c for $m > 0$. By **Proposition 4.6.18**, children of C can be classified as follows. Let

- term_c consist of those children $[s]_{\equiv_m^P}$ for which $(m-1)$ is terminal in π_s^P .
- For each possible priority dendrogram π , let $\text{mid}_{c,\pi}$ consist of those children $[s]_{\equiv_m^P}$ for which $(m-1)$ is nonterminal and $\text{spr}^{\pi_s^P}(m-1) = \pi$.

Every child of c belongs either to term_c or to exactly one $\text{mid}_{c,\pi}$. These classes describe precisely where the comparison relation between children can fail to be defined.

Proposition 4.6.19. *Let a and b be distinct $\equiv_m^P$ -equivalence classes of the same parent c , where $m > 0$. Then $\varepsilon_{a,b}^P = 0$ if and only if, for some priority dendrograms $\pi \neq \rho$, $a \in \text{mid}_\pi$ and $b \in \text{mid}_\rho$.*

Proof. Suppose that $a = [s]_{\equiv_m^P}$ and $b = [t]_{\equiv_m^P}$. Then we have $\mathbf{p}^P(s, t) = m$. Hence, by **Proposition 4.6.12**, $\varepsilon_{s,t}^P = 0$ if and only if $(m-1)$ is nonterminal in both π_s^P and π_t^P and

$$\text{spr}^{\pi_s^P}(m-1) \neq \text{spr}^{\pi_t^P}(m-1).$$

By the definition of term_c and $\text{mid}_{c,\pi}$, this is exactly the stated condition. $\square$

There is also a fixed direction whenever one child is terminal at the relevant node, and the other is nonterminal.

Proposition 4.6.20. *Let a and b be distinct $\equiv_m^P$ -equivalence classes of the same parent c , where $m > 0$. If $a \in \text{mid}_{c,\pi}$ and $b \in \text{term}_c$, then $\varepsilon_{a,b}^P = +1$.*

Proof. Again, let $a = [s]_{\equiv_m^P}$ and $b = [t]_{\equiv_m^P}$, so $\mathbf{p}^P(s, t) = m$. $(m-1)$ is nonterminal in π_s^P and terminal in π_t^P . By **Proposition 4.6.12**, we have an m -congruent ostensible 1-arity diagram $\mathbb{A}$ for s and t . In its completion $\mathbb{A}^*$, if x is the common image of the $(m-1)$ -th nodes before completion, then the nonterminal occurrence is represented by x , while the terminal occurrence is represented by x^* , and $\mathbb{A}(0 \vee 1)^* \models x < x^*$. The

comparison theorem therefore yields $\varepsilon_{s,t}^P = +1$. Since ε is invariant under the choice of representatives, the result follows. $\square$

For distinct children a and b of the common parent for which $\varepsilon_{a,b}^P \neq 0$, we define

$$a < b \iff \varepsilon_{a,b}^P = +1.$$

The preceding propositions show that $<$ is defined on every pair of children belonging to $\text{term}_c \cup \text{mid}_{c,\pi}$ for any fixed π . It turns out that $<$ is linear over this set.

Proposition 4.6.21. *Let c be an $\equiv_{m-1}^P$ -equivalence class, where $m > 0$. For every priority dendrogram π , the relation $<$ is a strict linear order on $\text{term}_c \cup \text{mid}_{c,\pi}$. Moreover, every element of $\text{mid}_{c,\pi}$ precedes every element of term_c .*

Proof. By [Proposition 4.6.19](#), any two distinct elements a, b of $\text{term}_c \cup \text{mid}_{c,\pi}$ have nonzero ε , so exactly one of $a < b$ or $b < a$ holds. It remains to verify transitivity. Suppose that $a, b, d \in \text{term}_c \cup \text{mid}_{c,\pi}$ and $a < b < d$. By [Proposition 4.6.14](#), we have $\varepsilon_{a,d}^P \in \{0, +1\}$. However, we can exclude the case $\varepsilon_{a,d}^P = 0$ by [Proposition 4.6.19](#), so we have $a < d$. Finally, [Proposition 4.6.20](#) gives $a < b$ for $a \in \text{mid}_{c,\pi}$ and $b \in \text{term}_c$. $\square$

4.7 2-Predendrograms

We finish this chapter with the 2-predendrograms, which must have formed a separate chapter.

Definition 4.7.1. A 2-predendrogram is a structure $(P, <, \multimap, \pi)$ satisfying the following:

1. $(P, \multimap)$ is a forest with the immediate successor relation $\multimap$. Moreover, if $\multimap^*$ is the transitive closure of $\multimap$, then for each $x \in P$, $\{y \in P \mid y \multimap^* x\}$ is finite and well-ordered by $\multimap^*$. We call the size of $\{y \in P \mid y \multimap^* x\}$ the *length* or *level* of x and denote it by $\text{lh}(x)$.

2. For every $x \in P$, either x is a terminal point (i.e., x has no immediate successor) or there is y such that $x \rightarrow^* y$ and y is a terminal point.
3. The *priority (1-)predendrogram* π is a function of domain P . For each $x \in P$, $\pi(x) = (\text{lh}(x), <_{\pi(x)})$ is a priority predendrogram of field $\text{lh}(x)$. Moreover,
 - $x \rightarrow y$ implies $\pi(x) \subseteq \pi(y)$.
 - If x and y have the same immediate predecessor, then $\pi(x) = \pi(y)$. For a non-terminal $x \in P$, we denote by $\text{spr}(x)$ the common priority predendrogram $\pi(y)$ of the immediate successors y of x , which we will call the *successor priority* of x .
4. The *local comparison relation* $\prec$ is a *binary relation* over P . Moreover, the following sets are linearly ordered by $\prec$:
 - The set of *roots* of P , i.e. elements with no immediate predecessor.
 - For a node $x \in P$ and a fixed priority dendrogram π , the set of $y \in P$ such that $P \models y \rightarrow x$, and either y is terminal or $\text{spr}(y) = \pi$.

Also, the two nodes are $\prec$ -incomparable when

- they have different levels,
- they have different immediate predecessors, or
- they are non-terminal and have different successor priority.

We also require that if both x and y have the same immediate predecessor, x is non-terminal, and y is terminal, then $x \prec y$.

Like the 1-predendrogram case, $\text{term}(P)$ denotes the set of terminal points of P . We also define $\text{pred}(x)$ as the $\rightarrow$ -increasing enumeration of $\{y \in P \mid y \rightarrow^* x\} \cup \{x\}$. $\text{pred}(x)$ is a sequence of length $\text{lh}(x) + 1$. If $\langle x_0, \dots, x_{m-1}, x_m \rangle$ is the sequence of predecessors of x and $l \leq m$, we write $x \upharpoonright l$ for x_l .

We can turn every 2-predendrogram into a 2-preptyx as follows:

Definition 4.7.2. For a 2-predendrogram P , let us define its *functorization* $\text{Fun}(P)$ as follows: For a 1-predendrogram C , we define $\text{Fun}(P)(C)$ by the set of

$$\langle x_0, c_0, x_1, \dots, x_{m-1}, c_{m-1}, x_m \rangle$$

such that $x_0, \dots, x_m \in P$ is a $\dashv$ -increasing chain with $\text{lh}^P(x_i) = i$, x_m is a terminal point of P , and $c_0, \dots, c_{m-1} \in C$ be such that the map $i \mapsto c_i$ is a 1-predendrogram morphism from $\pi^P(x_m)$ to D . We compare elements of $\text{Fun}(P)(C)$ by the Kleene-Brouwer order. For a 1-predendrogram morphism $f: C \rightarrow D$, we define $\text{Fun}(P)(f)$ by

$$\langle x_0, c_0, x_1, \dots, x_{m-1}, c_{m-1}, x_m \rangle \mapsto \langle x_0, f(c_0), x_1, \dots, x_{m-1}, f(c_{m-1}), x_m \rangle.$$

Although the local comparison relation $\prec$ of P is not necessarily a linear order on all siblings, the Kleene-Brouwer order is still linear. This is because if two immediate successors y, z of the same node are incomparable, then both are non-terminal and $\text{spr}(y) \neq \text{spr}(z)$. Hence two realizations through y and z cannot agree on all preceding parameter coordinates: otherwise the corresponding extensions of the common priority predendrogram would be simultaneously realized over the same prefix. Thus the Kleene-Brouwer comparison is decided in the parameter part before reaching the incomparable pair y, z .

We also define $\text{supp}_C^{\text{Fun}(P)}$ by

$$\langle x_0, c_0, x_1, \dots, x_{m-1}, c_{m-1}, x_m \rangle \mapsto \{c_0, c_1, \dots, c_{m-1}\} \subseteq D.$$

One can see that $\text{Fun}(P)$ is a Freund-style 2-preptyx. We can also define a cell decomposition of a 2-preptyx as follows.

Proposition 4.7.3. For a 2-preptyx P , let us define its cell decomposition $\text{Cell}(P)$ as follows:

1. $\text{Cell}(P) = \{[t]_{\equiv_n^P} \mid t \in \text{field } P \wedge n \leq |\text{arity } t|\}.$
2. $\text{Cell}(P) \models [s]_{\equiv_m^P} \multimap [t]_{\equiv_n^P} \text{ iff } n = m + 1 \text{ and } s \equiv_m^P t.$
3. $\text{Cell}(P) \models [s]_{\equiv_m^P} < [t]_{\equiv_n^P} \text{ iff } m = n, s \equiv_{m-1}^P t, \text{ and } \varepsilon_{s,t}^P = +1.$
4. $\pi^{\text{Cell}(P)}([s]_{\equiv_m^P}) = \pi^P(s) \upharpoonright m.$

For notational convenience, we assume that $s \equiv_{-1}^P t$ holds for every $s, t \in P$. Then $\text{Cell}(P)$ is a 2-predendrogram. $\text{Cell}(P)$ satisfies the following:

1. $[t]_{\equiv_m^P}$ is terminal iff $m = |\text{arity } t|.$
2. For a non-terminal $[t]_{\equiv_m^P}$, $\text{spr}^{\text{Cell}(P)}([t]_{\equiv_m^P}) = \pi^P(t) \upharpoonright (m + 1).$

Proof. We first check that the definitions of $\multimap$, $<$, and π are well-defined. Recall that $s \equiv_m^P t$ iff $s = t$ or $\mathbf{p}^P(s, t) > m$. Hence $\equiv_m^P$ refines $\equiv_{m-1}^P$, so $\multimap$ is well-defined. The local comparison relation is also well-defined by [Proposition 4.6.16](#), and π is well-defined by [Proposition 4.6.17](#).

We now verify the axioms of a 2-predendrogram. First, $(\text{Cell}(P), \multimap)$ is a forest. Its roots are precisely the $\equiv_0^P$ -classes, and for $m > 0$ the predecessors of $[s]_{\equiv_m^P}$ are

$$[s]_{\equiv_0^P} \multimap [s]_{\equiv_1^P} \multimap \cdots \multimap [s]_{\equiv_m^P}.$$

In particular, every node has a finite linearly ordered set of predecessors and $\text{lh}^{\text{Cell}(P)}([s]_{\equiv_m^P}) = m$. Every node also has a terminal extension since $[s]_{\equiv_m^P}$ is a terminal when $m = |\text{arity } s|$.

Next we verify the priority-predendrogram conditions. Clearly $\pi^{\text{Cell}(P)}([s]_{\equiv_m^P}) = \pi_s^P \upharpoonright m$ is a priority predendrogram of field m . Now we claim that two nodes of the same immediate predecessor have the same priority dendrogram. Suppose that $[s]_{\equiv_m^P} = [t]_{\equiv_m^P}$, so $s \equiv_m^P t$. By [Proposition 4.6.17](#), we have $\pi_s^P \upharpoonright (m + 1) = \pi_t^P \upharpoonright (m + 1)$, and so

$$\pi^{\text{Cell}(P)}([s]_{\equiv_{m+1}^P}) := \pi_s^P \upharpoonright (m + 1) = \pi^{\text{Cell}(P)}([t]_{\equiv_{m+1}^P}) := \pi_t^P \upharpoonright (m + 1).$$

It remains to verify the local comparison conditions. The first item of [Proposition 4.6.12](#) immediately implies that the set of roots is $\leq$ -comparable in $\text{Cell}(P)$. Moreover, the second item of [Proposition 4.6.12](#), [Proposition 4.6.18](#), [Proposition 4.6.19](#), and [Proposition 4.6.21](#) characterize when the two nodes are $\leq$ -comparable. [Proposition 4.6.14](#) also implies that $\leq$ is a linear order over the sets where two elements of the set become $\leq$ -comparable. Hence $\text{Cell}(P)$ is a 2-predendrogram. $\square$

Proposition 4.7.4. *For a predilator morphism $\iota: P \rightarrow Q$, if we define $\text{Cell}(\iota)$ by*

$$\text{Cell}(\iota)([s]_{\equiv_m^P}) = [\iota(s)]_{\equiv_m^Q},$$

then $\text{Cell}(\iota)$ is a well-defined 2-predendrogram morphism.

Proof. A 2-preptyx morphism $\iota: P \rightarrow Q$ preserves arities, priority predendrograms, $\mathbf{p}$, and ε . Hence $s \equiv_m^P t$ iff $\iota(s) \equiv_m^Q \iota(t)$, so $\text{Cell}(\iota)$ is well-defined and preserves the immediate successor relation. Preservation of ε gives preservation of $\leq$, while preservation of priority predendrograms gives

$$\pi^{\text{Cell}(Q)}([\iota(s)]_{\equiv_m^Q}) = \pi^{\text{Cell}(P)}([s]_{\equiv_m^P}).$$

Thus $\text{Cell}(\iota)$ is a 2-predendrogram morphism. $\square$

Cell and Fun are naturally isomorphic in the sense that a version of [Proposition 3.1.14](#) and [Proposition 3.1.16](#) holds. **Maybe add something here?**

Example 4.7.5. Let us calculate the cell decomposition of the evaluation ptyx $P: D \mapsto D(3)$. We can describe its terms explicitly. For a finite sequence p over $3 = \{0, 1, 2\}$ with no repetition, let τ_p denote the term which evaluates a $(\text{lh } p)$ -ary dilator term by substituting its arguments in priority order. We can apply a dilator term t to τ_p only when t is $(\text{lh } p)$ -ary, and the priority order of its arguments agrees with the relative order of the entries of p . More precisely, for every $i, j < \text{lh } p$, the i th argument precedes the j th argument in the priority order of t iff $p_i < p_j$. Hence we have

1. $\tau_\emptyset$ has arity $\text{Prime}_\emptyset$, which is the unique prime dilator with a nullary term.
2. τ_0, τ_1, τ_2 have arity Prime_0 , which is the unique prime dilator with a unary term.
3. $\tau_{01}, \tau_{02}, \tau_{12}$ have arity Prime_{01} , and $\tau_{10}, \tau_{20}, \tau_{21}$ have arity Prime_{10} .
4. For p with length 3, τ_p has arity Prime_p .

Now we calculate the term distance. Note that $\mathbf{p}^P(\tau_p, \tau_p) = |\pi_{\tau_p}^P| = \text{lh}(p) + 1$. Now we claim the following: let $k \leq \min(\text{lh } p, \text{lh } q)$ be the maximal number such that $p \upharpoonright k = q \upharpoonright k$, i.e., the first k entries of p and q agree. We claim that $\mathbf{p}^P(\tau_p, \tau_q) = k + 1$. If $m \leq k$, then consider the m -congruent 1-arity diagram $\mathbb{D}$ of τ_p and τ_q obtained by identifying the first m levels, differentiating the m th level, and putting the m th level for the arity of τ_p on the left, and the other on the right. We similarly obtain the m -congruent 1-arity diagram $\mathbb{K}$, but swapping the node order at level m . Then we have $P \models \tau_p <_{\mathbb{D}} \tau_q \wedge \tau_p >_{\mathbb{K}} \tau_q$, so $\mathbf{p}^P(\tau_p, \tau_q) > m$.

Hence we have $\mathbf{p}^P(\tau_p, \tau_q) \geq k + 1$. We now prove the equality:

- Suppose first that $k < \text{lh } p$ and $k < \text{lh } q$. By the maximality of k , we have $p(k) \neq q(k)$. Consider a $(k + 1)$ -congruent 1-arity diagram for τ_p and τ_q . (If there is no such 1-arity diagram, $\mathbf{p}^P(\tau_p, \tau_q) = k + 1$ follows immediately.) The first $k + 1$ nodes of the two priority predendrograms are identified, and hence the evaluations agree up to the k th node. The next comparison is therefore between the parameters $p(k)$ and $q(k)$. Since these are distinct elements of 3, their direction is independent of the choice of the $(k + 1)$ -congruent diagram. Hence $k + 1$ is secure.
- Suppose next that one of p and q is a proper initial segment of the other; without loss of generality, $\text{lh } p = k < \text{lh } q$. Then the node of index k is terminal in $\pi_{\tau_p}^P$ and non-terminal in $\pi_{\tau_q}^P$. Thus, in every $(k + 1)$ -congruent ostensible arity diagram, completion separates the two occurrences of the common node into its non-terminal copy and its terminal copy, whose local order is fixed. Consequently,

the comparison between τ_p and τ_q is independent of the choice of the $(k + 1)$ -congruent ostensible diagram. Hence $k + 1$ is secure also in this case.

From this fact, we can compute the cells:

1. There is only one cell in Level 0.
2. In Level 1, there are four cells: $\{\tau_\emptyset\}$ and the three sets C_0, C_1, C_2 , where $C_i = \{\tau_p \mid p(0) = i\}$.
3. In Level 2, each C_i has three children, namely, $\{\tau_i\}$ and C_{ij} for $j \neq i$, where $C_{ij} = \{\tau_p \mid p(0) = i, p(1) = j\}$.
4. In Level 3, C_{ij} has two children, each is $\{\tau_{ij}\}$ and another cell with the unique child of Level 4.

We also have the following:

1. $C_0 < C_1 < C_2 < \{\tau_\emptyset\}$,
2. $C_{ij} < \{\tau_i\}$.
3. The children C_{10} and C_{12} of C_1 are incomparable. All other children of other nodes are $\leq$ -comparable.

Figure 4.7 illustrates the cell decomposition of P . In this illustration, the order between C_{10} and C_{12} does *not* mean that they are $\leq$ -comparable. Otherwise, the order of children represents the $\leq$ -order. We use the isomorphic dilator instead of the priority dendrogram in Figure 4.7 since in this case, the dilator structure determines the unique priority dendrogram.

We now adapt the argument of Section 3.2 to characterize 2-ptykes in terms of 2-dendrograms. However, there are additional difficulties in the higher-type setting. For example, the union of priority dendrograms through an infinite branch may no longer be a predendrogram, since it may end up with an infinite chain with no nodes extending to

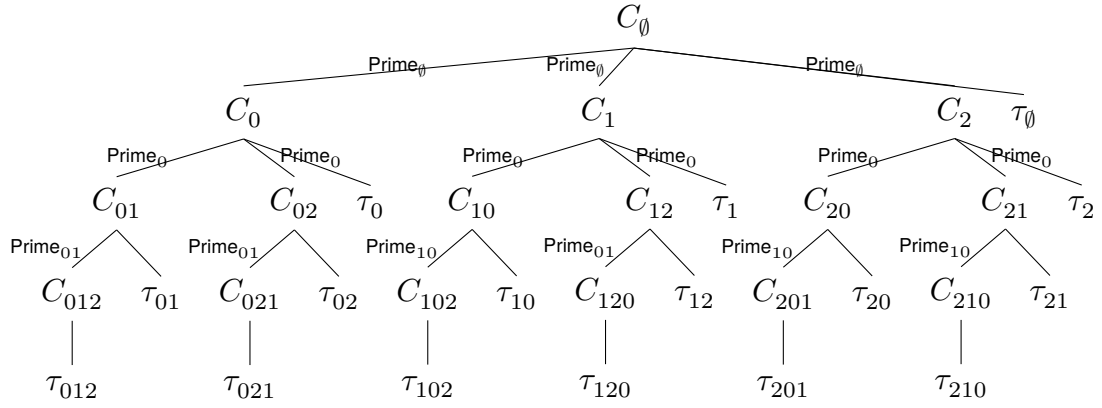


Figure 4.7: The cell decomposition of the evaluation ptyx $D \mapsto D(3)$.

a terminal node. Moreover, children of a single node may no longer be $\ll$ -comparable. Hence, we need a different formulation of thread and local well-foundedness. To handle the issue of the infinite branch, we distinguish the direct union along a branch from its completion. The former records the priority information appearing at finite stages, while the latter restores the terminal points which may disappear when passing to the union. More precisely, if $b = \langle x_n \mid n \in \mathbb{N} \rangle$ is an infinite branch of a 2-predendrogram P . We have $\pi^P(x_n) \subseteq \pi^P(x_{n+1})$ for every n , but we warn that the inclusion map is not a 2-predendrogram morphism. We only know that the inclusion map is a quasi-morphism. However, after taking completions, we obtain predendrogram morphisms $\iota_n^*: \pi^P(x_n)^* \rightarrow \pi^P(x_{n+1})^*$. This allows us to form a genuine predendrogram from the branch by taking the direct limit of the completed finite stages.

Definition 4.7.6. Let $b = \langle x_n \mid n \in \mathbb{N} \rangle$ be an infinite branch of a 2-predendrogram P . We define the *raw thread through b* as the union $\bigcup_{n < \omega} \pi^P(x_n)$. We also define the *thread predendrogram through b* by the direct limit $\lim_{\rightarrow n < \omega} \langle \pi^P(x_n), \iota_{n,m}^* \rangle$, where $\iota_{n,n+1}^*$ is the completion of the inclusion map $\pi^P(x_n) \subseteq \pi^P(x_{n+1})$, and

$$\iota_{n,m}^* = \iota_{m-1,m}^* \circ \cdots \circ \iota_{n,n+1}^*.$$

Calculating the thread predendrogram is not entirely trivial from the definition, since $\iota_{n,m}^*$ between the completed finite stages need not be inclusions. Suppose that a node x is terminal in $\pi^P(x_n)$, but not in $\pi^P(x_{n+1})$. Then the morphism $\iota_{n,n+1}^*$ sends x to the hidden terminal copy x^* , while x itself occurs in $\pi^P(x_{n+1})$ as a non-terminal node. Thus the direct limit cannot in general be obtained by simply taking the union of the finite completions. However, it turns out that we can compute the thread predendrogram by ‘completing’ the raw thread.

Proposition 4.7.7. *Let P be a 2-dendrogram, and $b = \langle x_n \mid n \in \mathbb{N} \rangle$ be an infinite branch of P . Let $U_b = \bigcup_{n < \omega} \pi^P(x_n)$, and U_b^* be the object obtained from U_b by applying the construction of ???. Then U_b^* is a predendrogram, and the thread predendrogram through b is isomorphic to U_b^* .*

Proof. Observe first that, once a node of the raw thread becomes non-terminal, it remains non-terminal at every later stage. Hence every $u \in U_b$ has one of the following two forms:

1. u is terminal in every sufficiently large $\pi^P(x_n)$, in which case u is terminal in U_b ;
2. u is non-terminal in every sufficiently large $\pi^P(x_n)$, in which case u is non-terminal in U_b .

We define a map $h: U_b^* \rightarrow \varinjlim_n \pi^P(x_n)^*$ as follows. If $u \in U_b$ is terminal in U_b , choose n with $u \in \pi^P(x_n)$ and send u to the direct-limit class $[n, u]$ represented by $u \in \pi^P(x_n)_n^*$. If u is non-terminal in U_b , choose n sufficiently large that u is non-terminal in $\pi^P(x_n)$, and send

$$u \longmapsto [n, u], \quad u^* \longmapsto [n, u^*].$$

These values do not depend on the choice of a sufficiently large n .

The only point requiring attention is the case where u is terminal at an earlier stage but non-terminal at a later one. If u is terminal in $\pi^P(x_n)$ and non-terminal in $\pi^P(x_{n+1})$,

then by the definition of the completion of a quasimorphism, $\iota_n^*(u) = u^*$. Thus the earlier terminal occurrence of u represents exactly the same direct-limit element as the hidden terminal copy u^* appearing at later stages. This is precisely the identification made in the definition of U_b^* .

It follows directly that h is bijective. Indeed, every element of the direct limit is represented at some finite stage either by a node $u \in \pi^P(x_n)$ or by its added terminal copy u^* , and the preceding discussion describes exactly which element of U_b^* it represents. Since the maps $\iota_{n,m}^*$ preserve the predecessor relation, the local comparison relation, terminal points, and priority predendrograms, the map h preserves all the corresponding relations. Hence h is a predendrogram isomorphism. Since the direct limit is a predendrogram, so is U_b^* . $\square$

We also need a new definition of local well-foundedness. Fortunately, the failure of the local comparison relation to be linear is rather limited. For each non-terminal node $x \in P$, there are only finitely many possible successor priorities among the immediate successors of x . Indeed, if y is an immediate successor of x , then $\text{spr}(y)$, whenever it is defined, is a finite priority predendrogram on a fixed finite field. Hence the immediate successors of x split into finitely many comparison blocks

$$\text{term}_x, \text{mid}_{x,\pi_0}, \dots, \text{mid}_{x,\pi_{k-1}},$$

where term_x consists of the successors whose relevant priority node is terminal and $\text{mid}_{x,\pi}$ consists of those whose relevant priority node is non-terminal with successor priority π . Recall that, for each π , $\text{term}_x \cup \text{mid}_{x,\pi}$ is linearly ordered by $<$, every element of $\text{mid}_{x,\pi}$ precedes every element of term_x , and elements belonging to distinct sets $\text{mid}_{x,\pi}$ and $\text{mid}_{x,\rho}$, for $\pi \neq \rho$, are incomparable.

Definition 4.7.8. A 2-predendrogram P is *locally well-founded* if

1. the set of roots of P is well-ordered by $<$, and

2. for every $x \in P$ and every successor priority π occurring among the immediate successors of x , $\text{term}_x \cup \text{mid}_{x,\pi}$ is well-ordered by $\leq$.

Equivalently, P is locally well-founded if the local comparison relation at every node has no infinite descending chain. Indeed, any descending chain of immediate successors can meet the terminal block at most initially, and once it enters a non-terminal block $\text{mid}_{x,\pi}$, it must remain in that same block. Since only finitely many successor-priority blocks occur, this is equivalent to the condition above.

We define 2-dendrograms by following the definition of 1-dendrograms:

Definition 4.7.9. A 2-predendrogram P is a 2-dendrogram if P is locally well-founded, and every thread predendrogram of P is not a dendrogram.

The following theorem provides the 2-ptyx analogue of [Theorem 3.2.4](#):

Theorem 4.7.10. For a 2-predendrogram P , $\text{Fun}(P)$ is a 2-ptyx iff P is a 2-dendrogram.

We also break up the proof of [Theorem 4.7.10](#) into several parts. The following lemma guarantees the local well-foundedness of a cell decomposition of a 2-ptyx.

Lemma 4.7.11. Suppose that P is a 2-predendrogram such that $\text{Fun}(P)$ is a 2-ptyx. Then P is locally well-founded.

Proof. Suppose that P is not locally well-founded. We claim that $\text{Fun}(P)$ is not a 2-ptyx. We consider the three cases. If the set of roots of P is ill-founded, let us fix the infinite $\leq$ -descending sequence $\langle x_n \mid n < \omega \rangle$ and $y_n \multimap^* x_n$. Then take $C_n = 0$ if $y_n = x_n$, and $C_n = \pi^P(y_n)$ otherwise. Then $C = \sum_{n < \omega} C_n$ is a dendrogram since it has no infinite branching, the set of roots of C has ordertype ω , and the set of immediate successors of a fixed node of C is finite. However, $\text{Fun}(P)(C)$ is ill-founded.

Now suppose that the local well-foundedness fails for the set of children of $x \in P$. We have two possible cases: The set of terminal children term_x of x is ill-founded under $\leq$, or the set $\text{mid}_{x,\pi}$ is ill-founded under $\leq$ for some priority dendrogram π . In the former

case, $\text{Fun}(P)(\text{spr}^P(x))$ is ill-founded, yet $\text{spr}^P(x)$ is a (finite) dendrogram. In the latter case, let us fix a $\leq$ -infinite-descending sequence $\langle y_n \mid n < \omega \rangle$, and choose a terminal $z_n \prec^* y_n$. Then $\text{spr}^P(y_n) = \pi$ for every n , and π is a subtree of $\pi^P(z_n)$.

We first choose a subsequence so that we can amalgamate $\pi^P(z_n)$ compatibly over π . For each $u \in \pi$, record whether u remains terminal in $\pi^P(z_n)$, and if not, record its successor priority $\text{spr}^{\pi^P(z_n)}(u)$. There are only finitely many such records since π is finite. Hence, by the pigeonhole principle, we may assume that the records are all the same for every n . It follows that the inclusion maps $\pi \rightarrow \pi^P(z_n)$ and $\pi \rightarrow \pi^P(z_m)$ as quasimorphisms satisfy the compatibility condition of the predendrogram amalgamation, and whenever immediate successors are added above a common node of π , their priority predendrograms agree.

Then we construct finite predendrograms $A_0 \subseteq A_1 \subseteq A_2 \subseteq \dots$ and embeddings $e_n: \pi^P(z_n) \rightarrow A_n$. We take $A_0 = \pi^P(z_0)$ and e_0 to be the identity map. Once we constructed A_n , we construct A_{n+1} by amalgamating $e_n \upharpoonright \pi: \pi \rightarrow A_n$ and the inclusion map $\pi \rightarrow \pi^P(z_{n+1})$, in a way that we place new points coming from $\pi^P(z_{n+1})$ after the points already occurring in A_n . It gives the amalgamation maps $A_n \rightarrow A_{n+1}$ and $e_{n+1}: \pi^P(z_{n+1}) \rightarrow A_{n+1}$. Then take $A = \bigcup_{n < \omega} A_n$, where we identify each A_n with its image in A_{n+1} .

By construction, no new predecessor relation is introduced between points coming from different amalgamands outside the common copy of π . Hence, once a branch leaves π , it is contained in the image of some finite predendrogram $\pi^P(z_n)$. Thus A has no infinite branch. Moreover, A is locally well-founded. Indeed, at each amalgamation step, within every local comparison interval determined by the common copy of π , the new points are placed after the points already present. Hence every local order of A is an increasing union of finite linear orders obtained by adjoining new points only on the right, and is therefore a well-order. The same applies to the order on the roots. That is,

A is a dendrogram.

For each n , regard e_n as an embedding $e_n: \pi^P(z_n) \rightarrow A$. Let

$$\sigma_n = \langle z_n \upharpoonright 0, e_n(0), z_n \upharpoonright 1, e_n(1), \dots, e_n(\text{lh}(z_n) - 1), z_n \rangle \in \text{Fun}(P)(A).$$

The predecessor sequences of z_n and z_{n+1} agree up to x , and their next nodes are y_n and y_{n+1} , respectively. Moreover, the embeddings e_n agree on the common copy of π , so all parameter coordinates occurring before this first difference agree. Since $y_n > y_{n+1}$, the Kleene–Brouwer order gives $\sigma_n > \sigma_{n+1}$. Therefore, $\langle \sigma_n \mid n < \omega \rangle$ is an infinite descending sequence in $\text{Fun}(P)(A)$, contradicting that $\text{Fun}(P)$ is a 2-ptyx. $\square$

Proposition 4.7.12. *Suppose that P is a locally well-founded 2-predendrogram. If $\text{Fun}(P)(D)$ is ill-founded for some predendrogram D , then P has a thread dendrogram that is embedded into D^* . That is, we can find an infinite chain $\langle x_n \mid n < \omega \rangle$ of immediate successors in P such that $(\bigcup_{n < \omega} \pi_{x_n}^P)^*$ embeds to D^* .*

Proof. Fix an infinite descending sequence $\langle \sigma_n \mid n < \omega \rangle$ in $\text{Fun}(P)(D)$. Consider the tree T obtained from $\text{Fun}(P)(D)$ by taking the closure under initial segments of $\sigma \in \text{Fun}(P)(D)$. We claim that every set of immediate successors in T is well-ordered. Indeed, if an initial segment ends with a node of D , then its immediate successors are nodes of P lying in a single local comparison block of P , and hence are well-ordered since P is locally well-founded. If an initial segment ends with a node of P , then its immediate successors are nodes of D lying either among the roots or among the immediate successors of a fixed node of D , and hence are well-ordered since D is locally well-founded.

We can think of $\langle \sigma_n \mid n < \omega \rangle$ as an infinite descending sequence over T under the Kleene–Brouwer order. Applying the usual Kleene–Brouwer stabilization argument, we have that the length $|\sigma_n|$ is unbounded in n , and there are $x_0, x_1, x_2, \dots \in P$ and

$c_0, c_1, c_2, \dots \in D$ such that every finite initial segment of

$$\langle x_0, c_0, x_1, c_1, x_2, c_2, \dots \rangle$$

occurs as an initial segment of elements of σ_n arbitrarily far along the sequence. In particular, we also have $P \models x_0 \multimap x_1 \multimap x_2 \multimap \dots$.

Let $U = \bigcup_{n < \omega} \pi^P(x_n)$. Define $e: U \rightarrow D$ by $e(i) = c_i$. Every finite part of U is contained in some $\pi^P(x_n)$, and the corresponding initial segment of the branch of T extends to an element of $\text{Fun}(P)(D)$. Hence e preserves the predecessor relation, the local comparison relation, and priorities. It may fail to preserve terminal points, and this is the only possible failure. The completion construction for predendrogram quasimorphisms therefore applies verbatim and yields a predendrogram embedding $e^*: U^* \rightarrow D^*$. Since U^* is the thread predendrogram associated with $\langle x_n \mid n < \omega \rangle$, the desired thread predendrogram embeds into D^* . $\square$

Proposition 4.7.13. *Suppose that P is a 2-predendrogram such that $\text{Fun}(P)$ is a 2-ptyx. Then every thread predendrogram of P is not a dendrogram.*

Proof. Suppose that there is a thread predendrogram of P that is a dendrogram. That is, we can find an infinite sequence

$$P \models x_0 \multimap x_1 \multimap x_2 \multimap \dots$$

such that $U^* := (\bigcup_{m < \omega} \pi_{x_m}^P)^*$ is a dendrogram. For notational convenience, let us write $C_n = \pi^P(x_n)$.

For each $m < \omega$, choose a terminal extension $z_m \multimap^* x_m$. We first thin out the sequence of terminal extensions. For $n < m$, consider the following finite information about the inclusion $C_n \subseteq \pi^P(z_m)$: for each $u \in \pi^P(x_n)$, record whether u is terminal in $\pi^P(z_m)$, and, if it is non-terminal, record $\text{spr}^{\pi^P(z_m)}(u)$. We call this the C_n -profile

of z_m . There are only finitely many possible C_n -profiles since C_n is finite, and there are only finitely many possible successor priorities.

By applying the pigeonhole principle infinitely many times, we can choose infinite sets $I_0 \supseteq I_1 \supseteq I_2 \supseteq \dots$ such that $I_n \subseteq \omega \setminus (n+1)$ and the C_n -profile of z_m is independent of $m \in I_n$. Choose a strictly increasing sequence $k_n \in I_n$ and set $y_n = z_{k_n}$. Then y_n extends x_{n+1} , and hence $C_{n+1} \subseteq \pi^P(y_n)$. Moreover, $\pi^P(y_n)$ and $\pi^P(y_{n+1})$ have the same C_n -profile for every n . Now we prove the following claim by amalgamating $\pi^P(y_n)$ appropriately:

Claim. There is a dendrogram D and predendrogram embeddings $e_n: \pi^P(y_n) \rightarrow D$ such that for every $n < \omega$,

- $e_{n+1} \upharpoonright C_n = e_n \upharpoonright C_n$, and
- $D \models e_{n+1}(n) < e_n(n)$.

Once we have the claim, we have the desired result. For each $n < \omega$, let

$$\sigma_n = \langle y_n \upharpoonright 0, e_n(0), y_n \upharpoonright 1, e_n(1), \dots, y_n \rangle \in \text{Fun}(P)(D).$$

Since y_n and y_{n+1} both extend x_{n+1} , their predecessor sequences agree through x_{n+1} . Moreover, we have $e_n(i) = e_{n+1}(i)$ for every $i < n$, so σ_n and σ_{n+1} agree up to the n -th parameter coordinate. At the n -th parameter coordinate, $e_{n+1}(n) < e_n(n)$ gives $\sigma_n >_{\text{KB}} \sigma_{n+1}$. That is, $\langle \sigma_n \mid n < \omega \rangle$ is an infinite descending sequence of $\text{Fun}(P)(D)$, so $\text{Fun}(P)$ is not a 2-ptyx.

Now we prove the claim. We construct finite predendrograms $D_0 \subseteq D_1 \subseteq D_2 \subseteq \dots$ and embeddings $e_n: \pi^P(y_n) \rightarrow D_n$ recursively such that

1. $e_{n+1} \upharpoonright C_n = e_n \upharpoonright C_n$, and
2. $D_{n+1} \models e_{n+1}(n) < e_n(n)$ for every n .
3. The map $e_n \upharpoonright C_n: C_n \rightarrow D_n$ has in D_n the same terminality and successor-priority profile as the inclusion quasimorphism $C_n \subseteq \pi^P(y_n)$.

In the following argument, we identify D_n with its image in D_{n+1} . We first take $D_0 = \pi^P(y_0)$ and let e_0 be the identity map. Now suppose that D_n and e_n have been constructed. By our choice of the y_i 's, the inclusions $C_n \subseteq \pi^P(y_n)$ and $\pi^P(x_n) \subseteq \pi^P(y_{n+1})$ have the same profile. Hence

$$e_n \upharpoonright C_n : C_n \rightarrow D_n \quad \text{and} \quad C_n \subseteq \pi^P(y_{n+1})$$

satisfy the compatibility condition for predendrogram amalgamation. Moreover, their terminality agrees on C_n .

Now we are going to amalgamate D_n and $\pi^P(y_{n+1})$ over C_n to form the following commute diagram:

$$\begin{array}{ccc} C_n & \xrightarrow{e_n \upharpoonright C_n} & D_n \\ \downarrow \subseteq & & \downarrow \\ \pi^P(y_{n+1}) & \xrightarrow{e_{n+1}} & D_{n+1} \end{array}$$

Since both $\pi^P(y_n)$ and $\pi^P(y_{n+1})$ contain C_{n+1} , the two copies $e_n(n)$ and $e_{n+1}(n)$ have the same immediate predecessor and determine the same cut over the common copy of C_n . Hence we may choose the local comparison order in the amalgamation so that $e_{n+1}(n) \leq e_n(n)$.

We also choose the local comparison orders in the amalgamations in a coherent way. To explain the choice, fix a node u in the common copy of C_n , and suppose that the immediate successors of u which already belong to the common part are

$$a_0 \leq a_1 \leq \dots \leq a_{r-1}.$$

The immediate successors contributed by either amalgamand are divided into finite blocks according to the intervals

$$(-\infty, a_0), \quad (a_0, a_1), \quad \dots, \quad (a_{r-2}, a_{r-1}), \quad (a_{r-1}, +\infty)$$

in which they occur. Within each such block we keep the local order inherited from the original amalgamand. If finite blocks arising at different stages occupy the same interval,

we arrange these blocks in increasing order of the stage at which they are introduced. Thus, schematically, if B_i is introduced before B_j in the same interval, we arrange

$$a_k < B_i < B_j < a_{k+1},$$

while preserving the given finite orders inside B_i and B_j . We make the analogous choice for the order of roots. The distinguished pair $e_{n+1}(n), e_n(n)$ is arranged as prescribed above: $e_{n+1}(n) < e_n(n)$. The point $e_{n+1}(n)$ becomes part of the common copy at all later stages, whereas $e_n(n)$ belongs only to the finite piece introduced at stage n . Hence this exceptional insertion occurs only once for each n .

We use the amalgamation constructed in the proof of [Theorem 3.3.11](#), so that the images of D_n and $\pi^P(y_{n+1})$ intersect exactly in the common image of C_n . In particular, no points outside the common copy of C_n are identified. Moreover, as in the construction of the amalgamation, the predecessor relation is inherited from the two amalgamands; only the local comparison orders need to be chosen.

Finally, Condition (3) is preserved. Note that $C_{n+1} = C_n \cup \{n\}$. For $u \in C_n$, the two amalgamands have the same terminality and successor-priority profile at u . Hence the amalgamation does not change this profile. For the new node n , the two copies $e_n(n)$ and $e_{n+1}(n)$ have the same immediate predecessor (or are both roots), but $e_n(n)$ does not become a successor or predecessor of $e_{n+1}(n)$. Hence the amalgamation introduces no new immediate successor of $e_{n+1}(n)$, so its terminality and successor priority are exactly those inherited from $\pi^P(y_{n+1})$. Thus condition (3) holds at stage $n + 1$.

Now we claim that $D = \bigcup_{n < \omega} D_n$ is a dendrogram. The previous amalgamation construction guarantees that D is locally well-founded. Now we claim that every infinite branch is contained in $\overline{U} = \{\overline{u} \mid u \in U\}$, where $U = \bigcup_{n < \omega} C_n$ and $\overline{u}$ is the eventual value of $e_n(u)$, which exists since $e_{n+1} \upharpoonright C_n = e_n \upharpoonright C_n$. Indeed, if $v \in D_n \setminus e_n[C_n]$, then no later amalgamation adds any new extension above v . This is because the amalgamation at stage n is performed only over the common copy of C_n , and the predeces-

sor relation of the amalgam is simply inherited from the two amalgamands. Likewise, among the new points contributed by $\pi^P(y_{n+1})$ at stage $n + 1$, only the point belonging to $C_{n+1} \setminus C_n = \{n\}$ becomes part of the common copy used in later amalgamations. Every point of $\pi^P(y_{n+1}) \setminus e_{n+1}[C_{n+1}]$ belongs to $D_{n+1} \setminus e_{n+1}[C_{n+1}]$, and hence receives no new extensions at any later stage. Consequently, for every point $v \in D \setminus \overline{U}$, the subtree of D above v is contained in some finite D_n . Thus no infinite branch of D can pass through v , and every infinite branch of D is contained in $\overline{U}$. Since U^* is a dendrogram and D is well-founded, we have that D^* is a dendrogram, so is D as desired. $\square$

Proof of Theorem 4.7.10. Suppose that P is a 2-dendrogram. Then Proposition 4.7.12 implies $\text{Fun}(P)$ is a 2-ptyx. Conversely, if $\text{Fun}(P)$ is a 2-ptyx, then Lemma 4.7.11 implies P is locally well-founded, and Proposition 4.7.13 implies every thread predendrogram of P is not a dendrogram. Hence P is a 2-dendrogram. $\square$